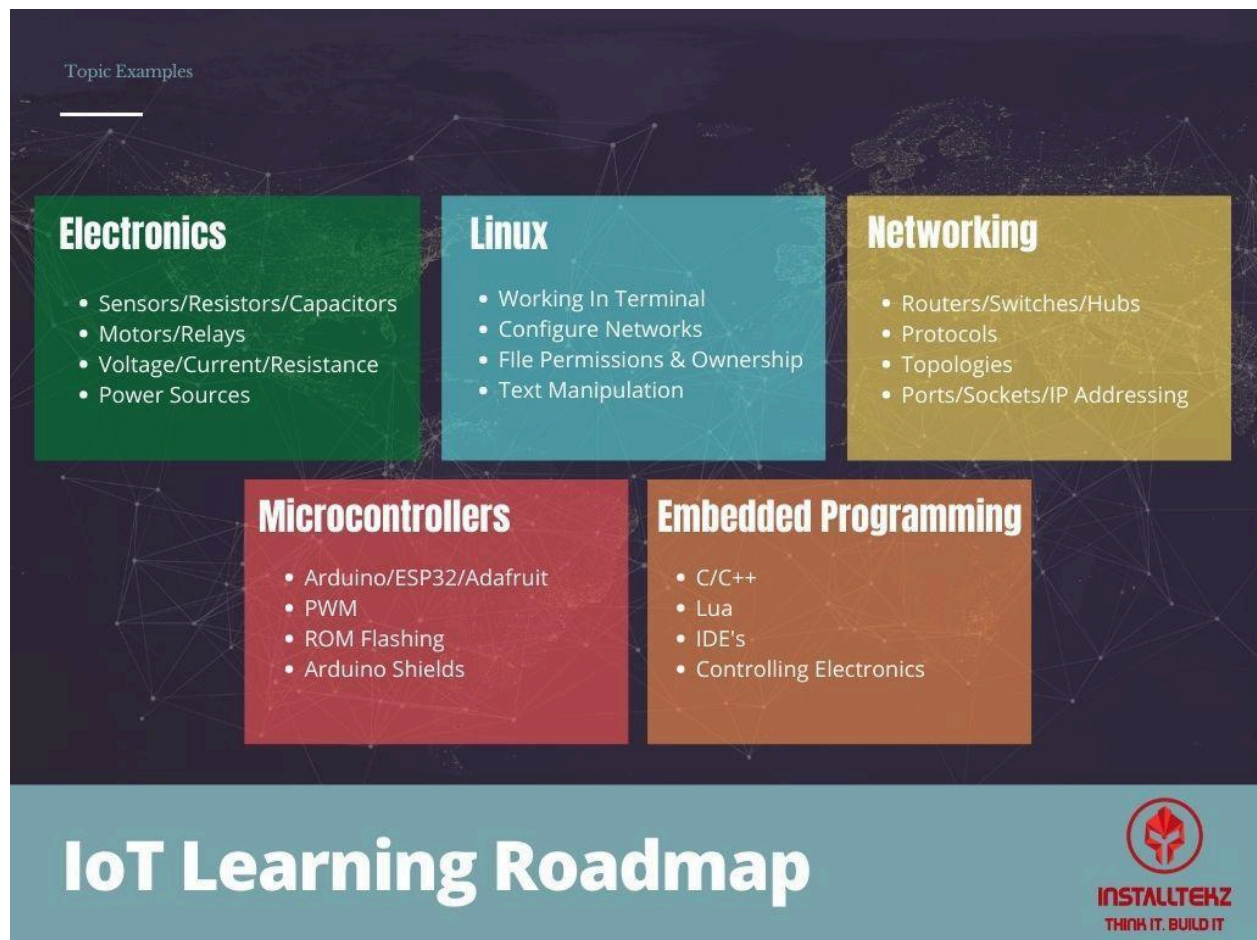
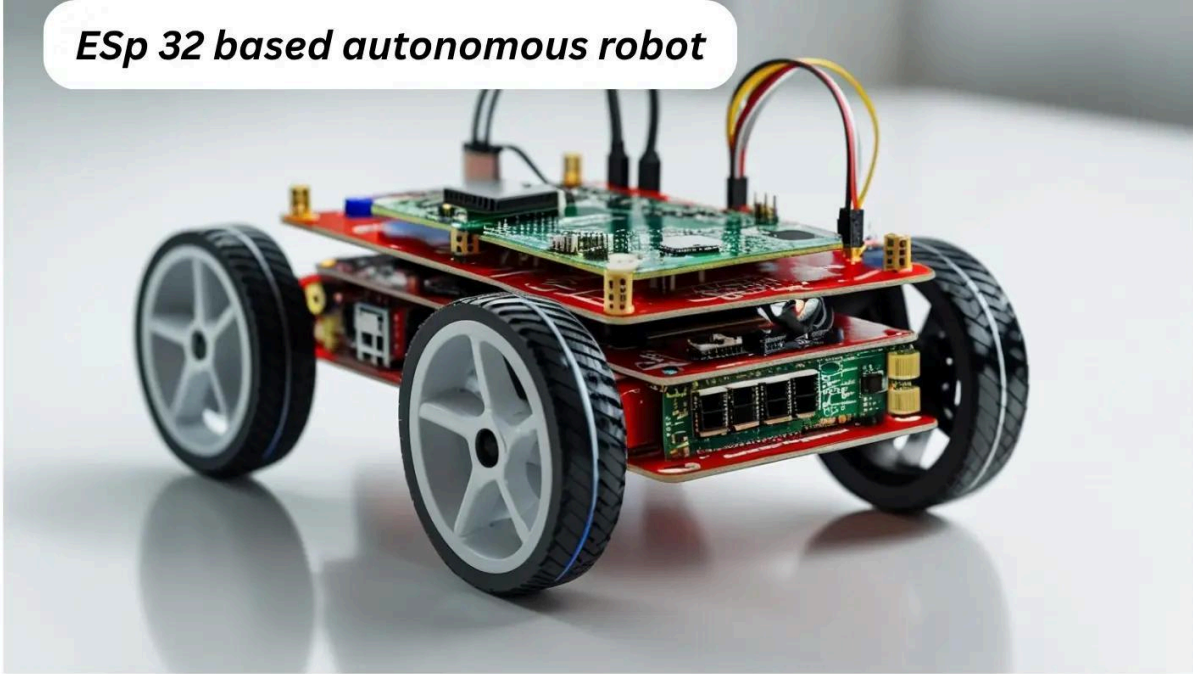


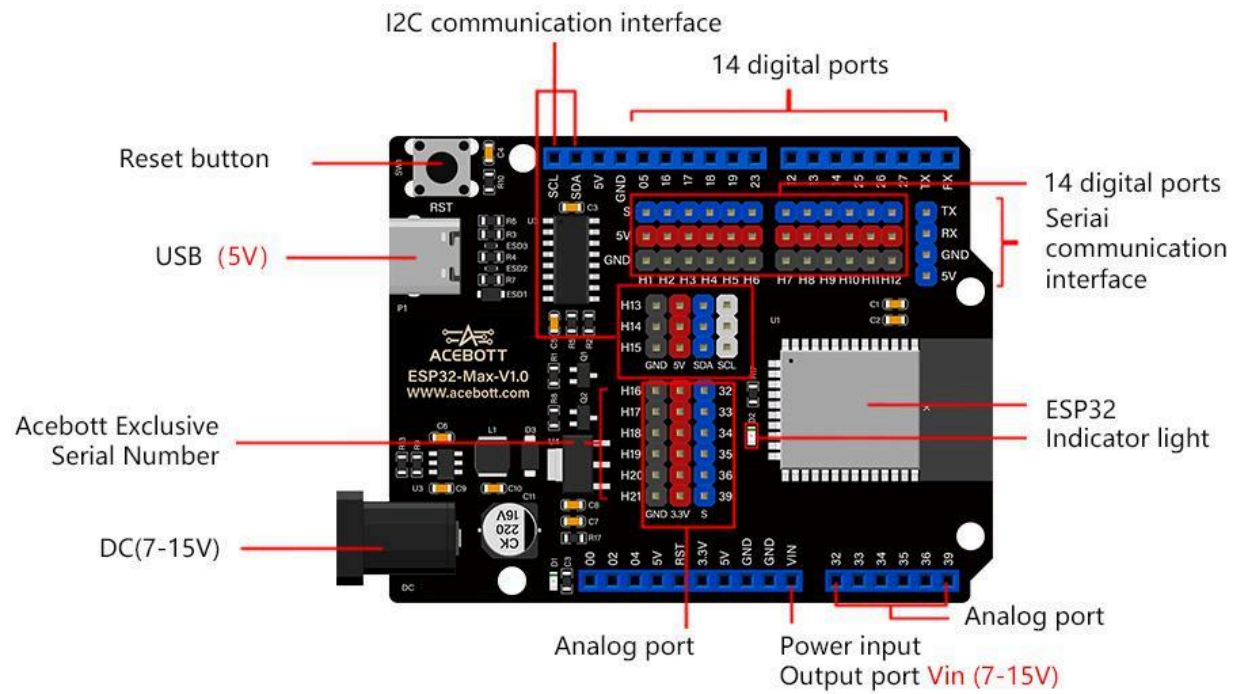
IoT + Embedded Systems Learning Roadmap



ESp 32 based autonomous robot

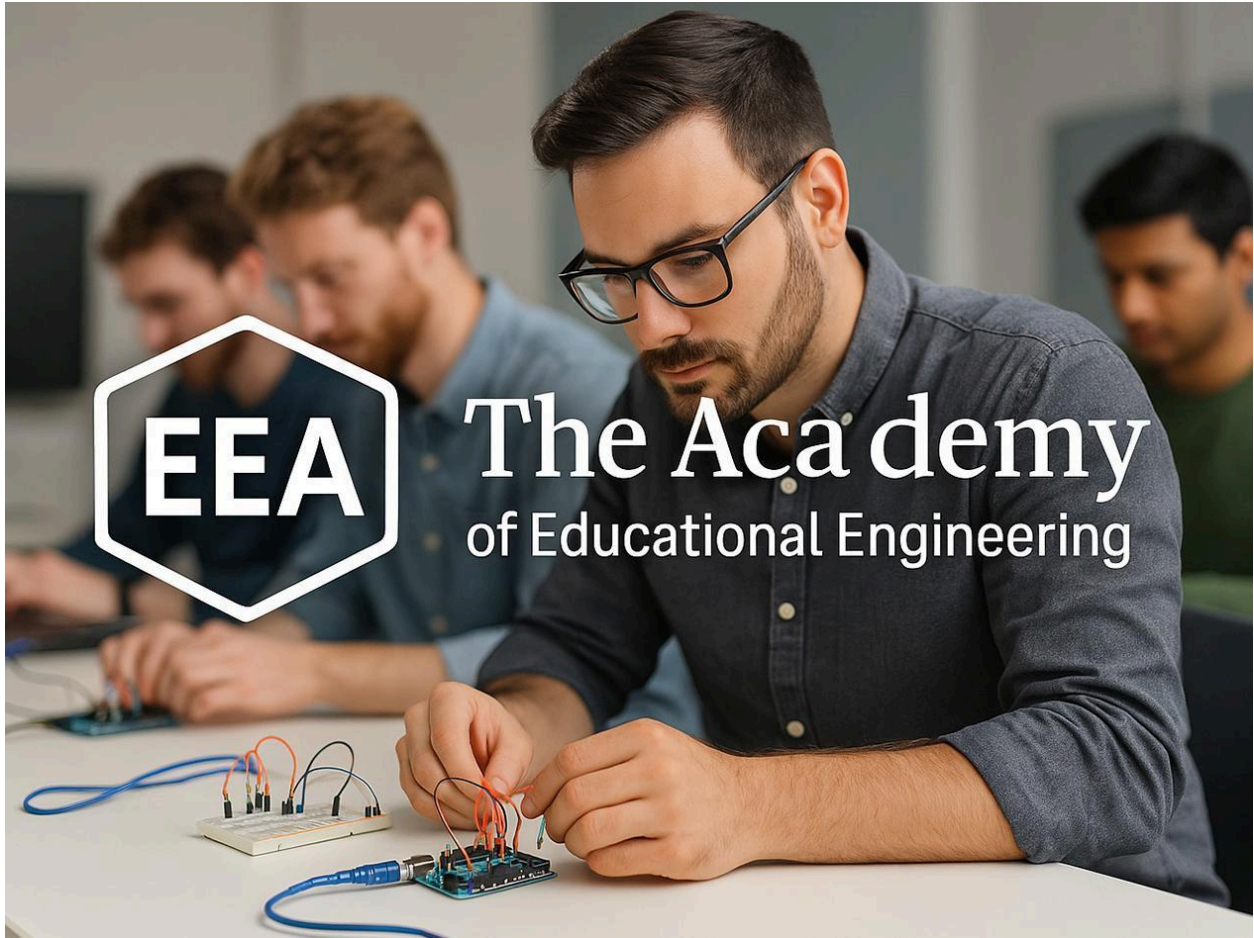


ESP32 Controller -Overview





The Academy of Educational Engineering



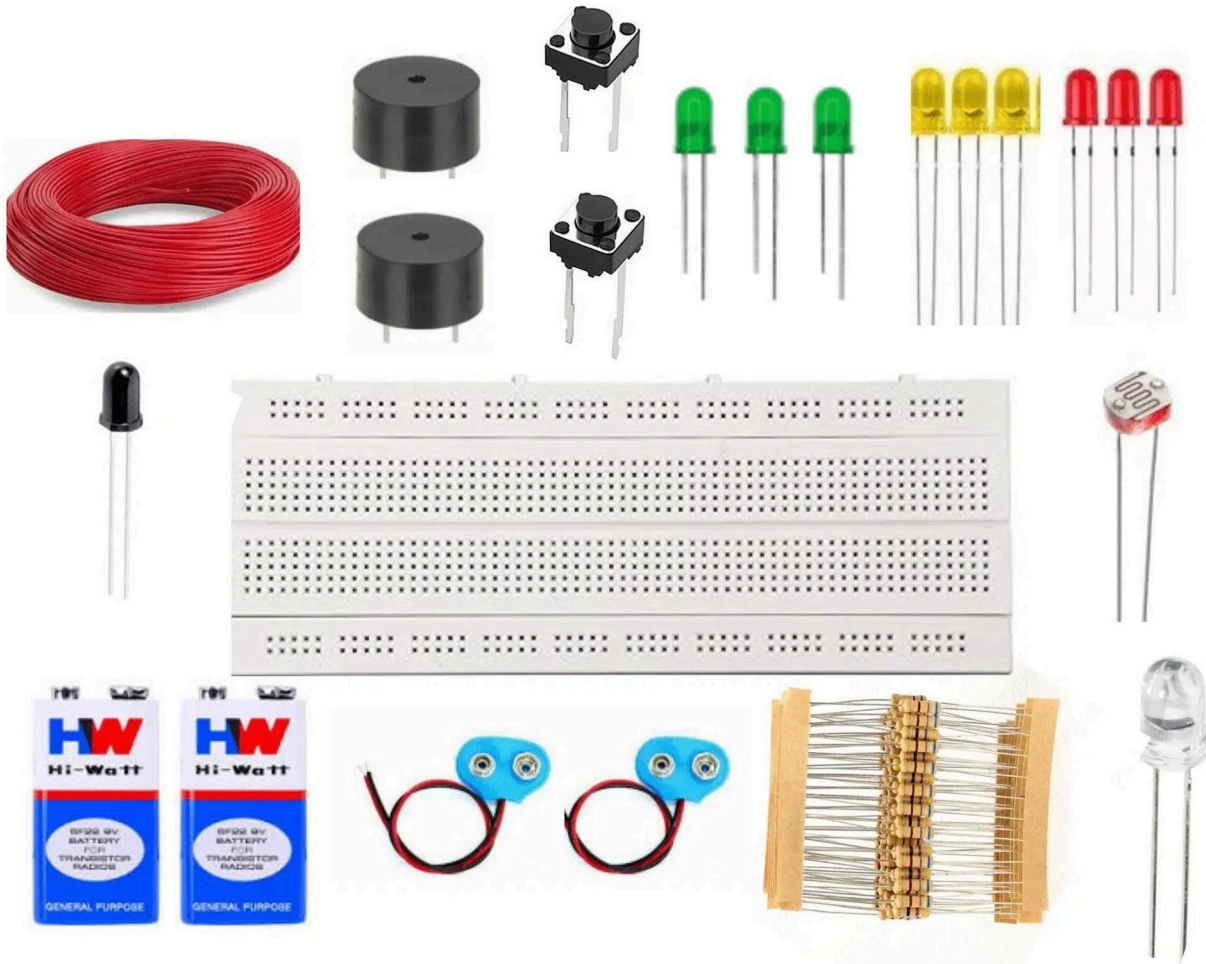


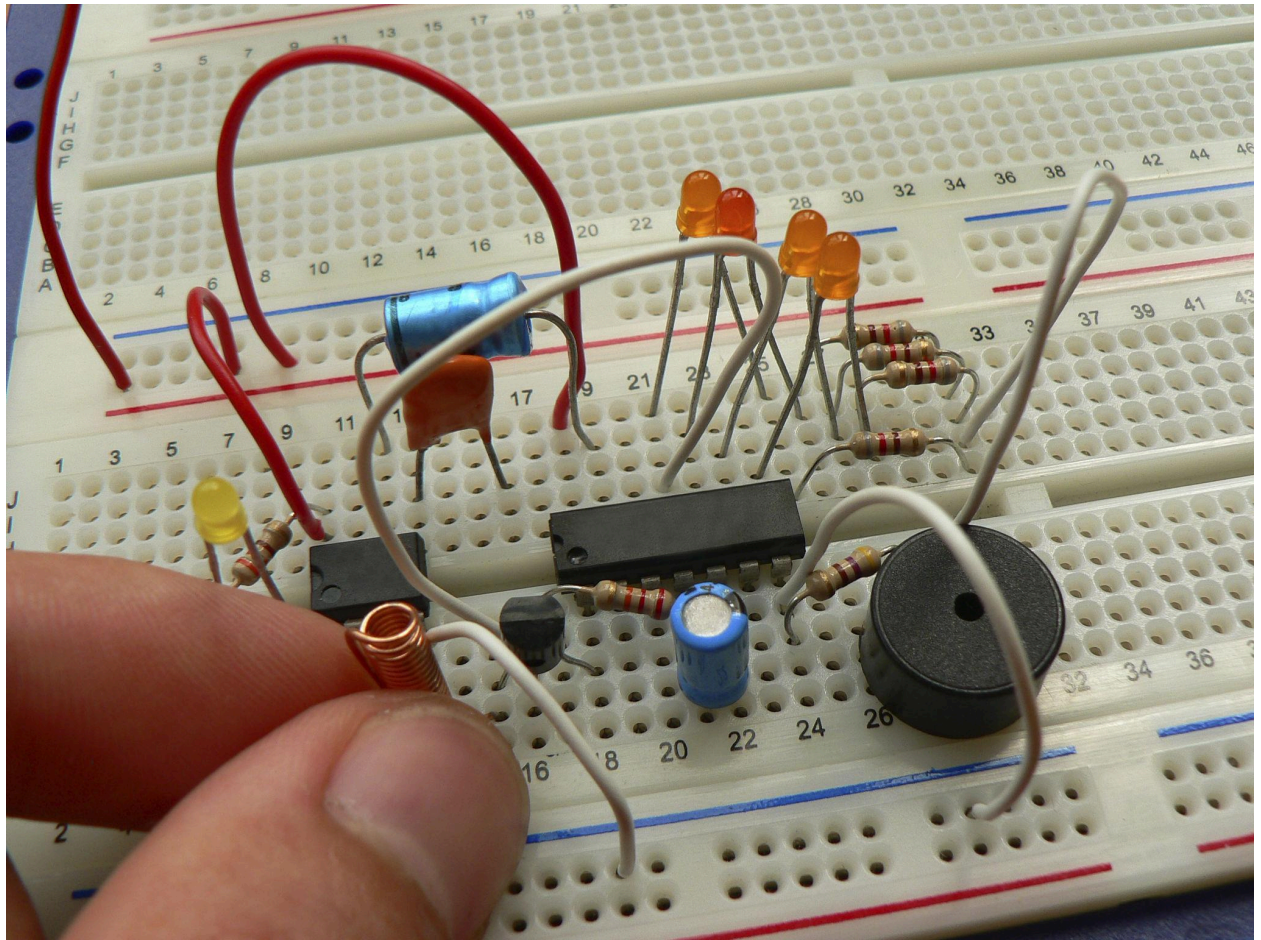
This roadmap takes you from beginner → advanced in:

- Embedded Systems
- IoT
- AI
- Robotics
- Automation

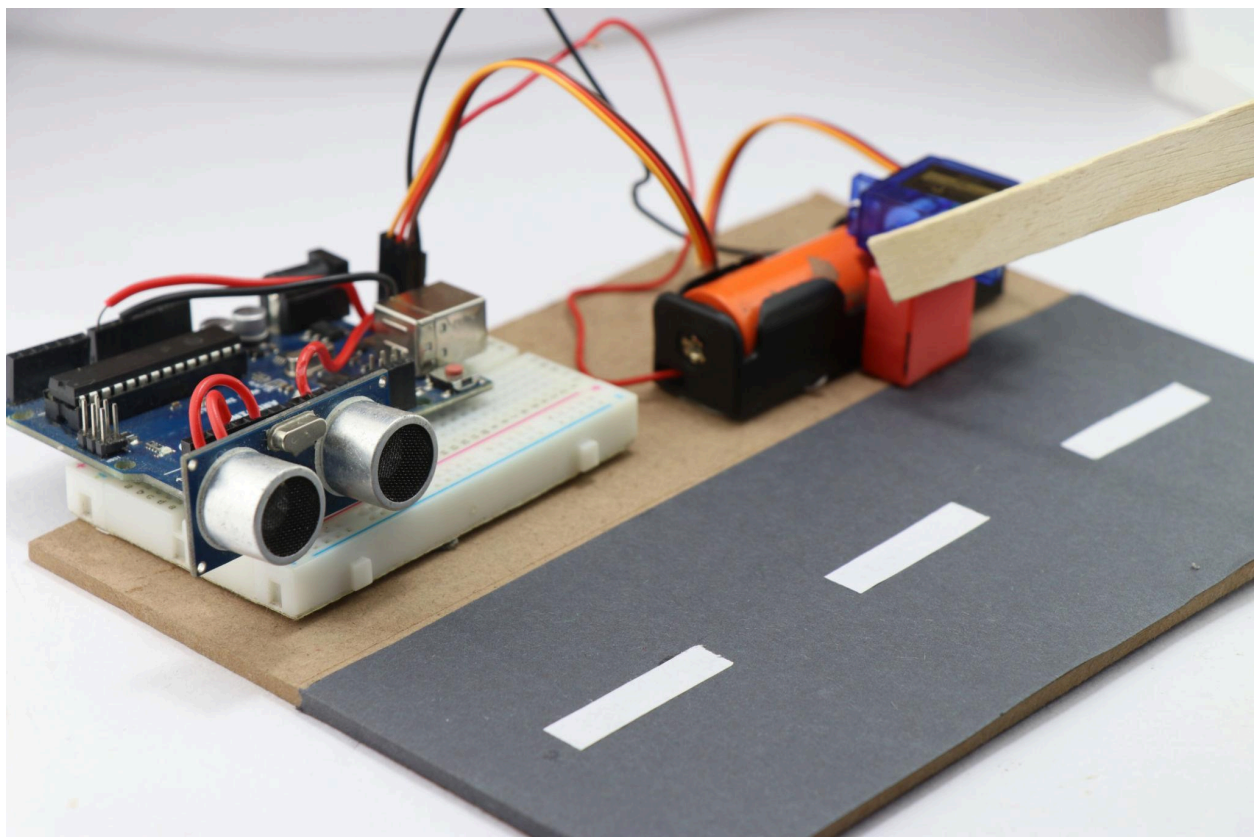
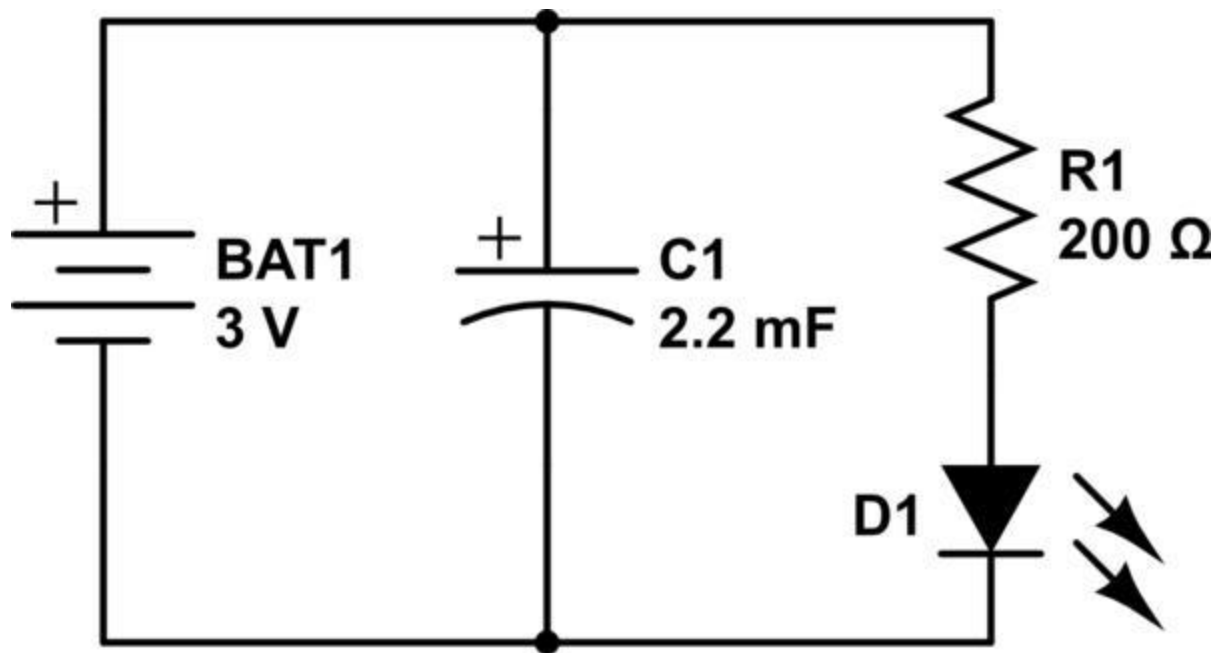
1. Electronics Basics

ELECTRONIC COMPONENT AND SENSING KIT









Learn

- Voltage
- Current
- Resistance
- Ohm's Law
- Series & parallel circuits

Components

- Resistor
- Capacitor
- LED
- Diode
- Transistor

Practice

- LED blinking
- Push button circuit
- Buzzer control

Tools

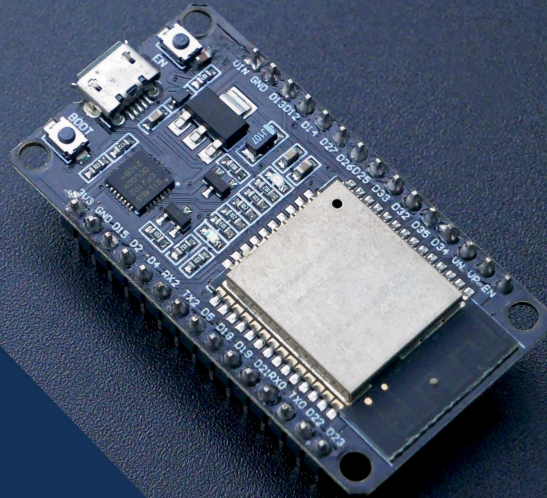
- Breadboard
 - Multimeter
 - Soldering kit
-

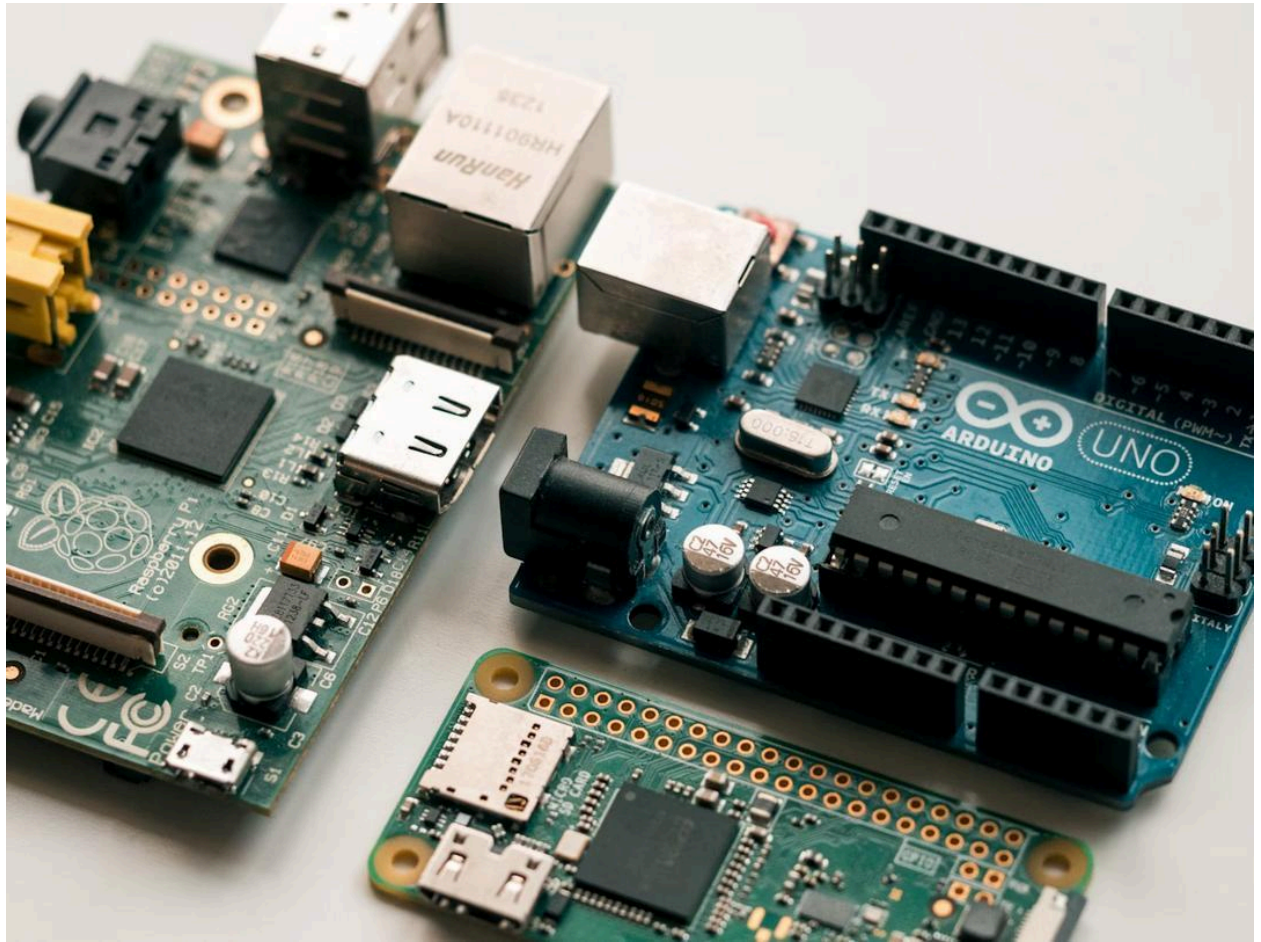
2. Arduino + ESP32

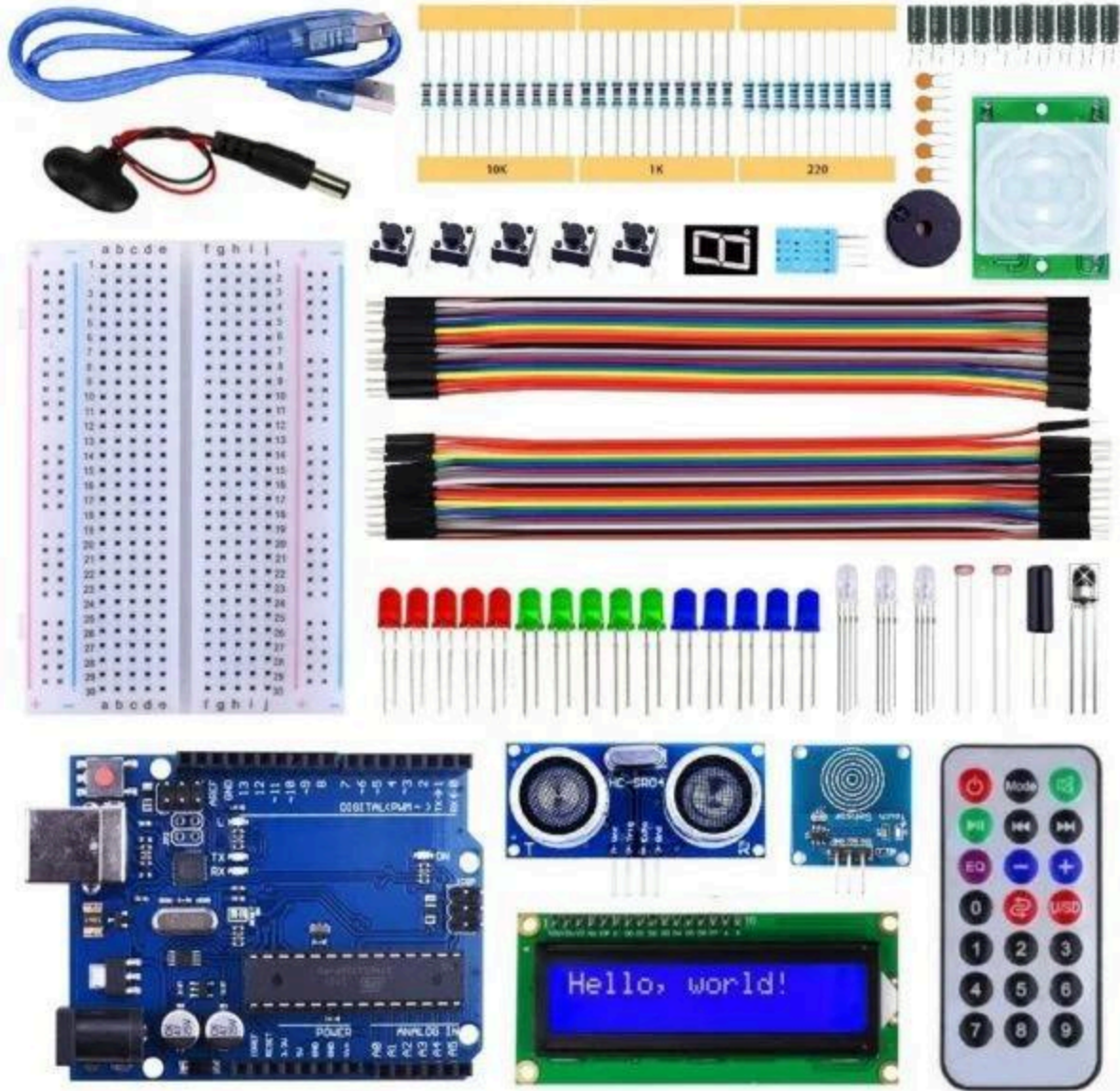
Getting Started with ESP32 Wi-Fi & Bluetooth SoC using DOIT ESP32 DevKit V1

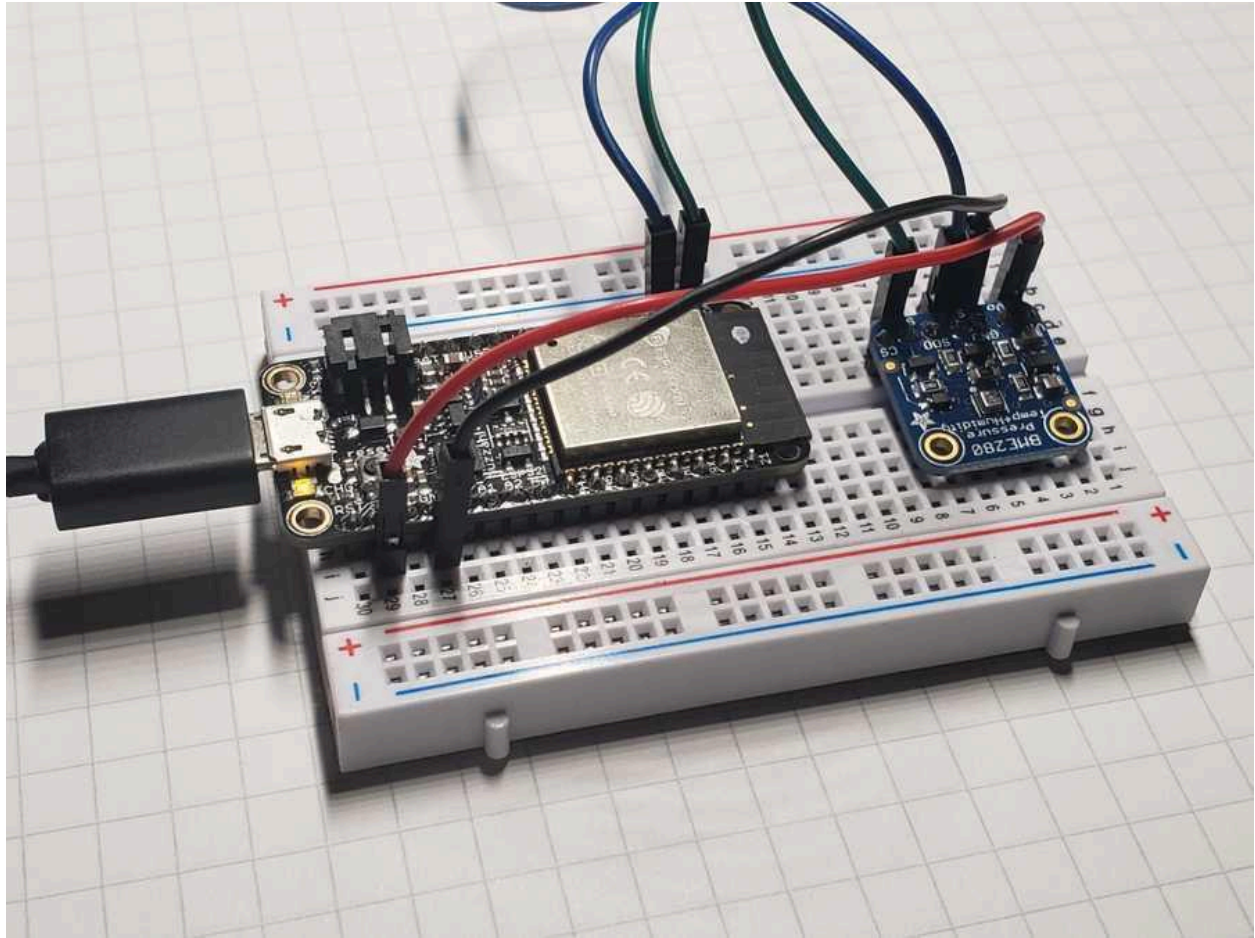
Learn how to use Espressif ESP32 SoC for Wi-Fi and Bluetooth development using DOIT ESP32 DevKit V1 development board. Use Arduino, ESP-IDF, PlatformIO and VS Code for software development.

www.circuitstate.com/getstartesp32

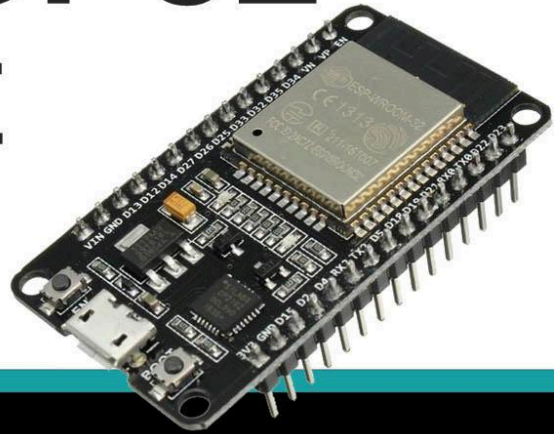




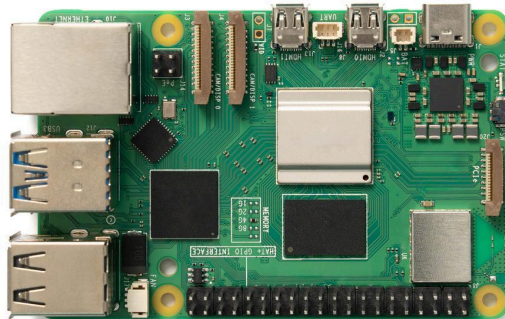




Install ESP32 in Arduino IDE 1 minute 🕒



Model: ESP32 (ESP32 Module), 50 MHz, Flash: Disabled, All: SSL, esp8266 (most compatible), 4M (no SPIFFS), v2: Lower Memory, Disabled, None, Only Sketch, 115200



Learn

- GPIO control
- PWM
- Serial communication
- Interrupts
- Timers

Projects

- LED control
- Servo motor
- Wi-Fi LED control
- OLED display

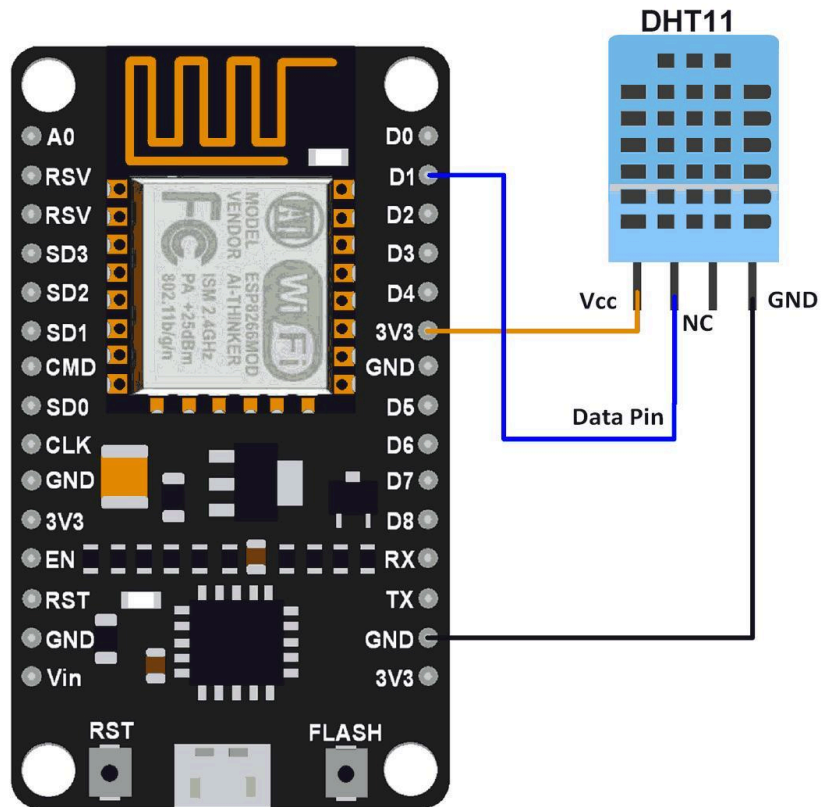
Languages

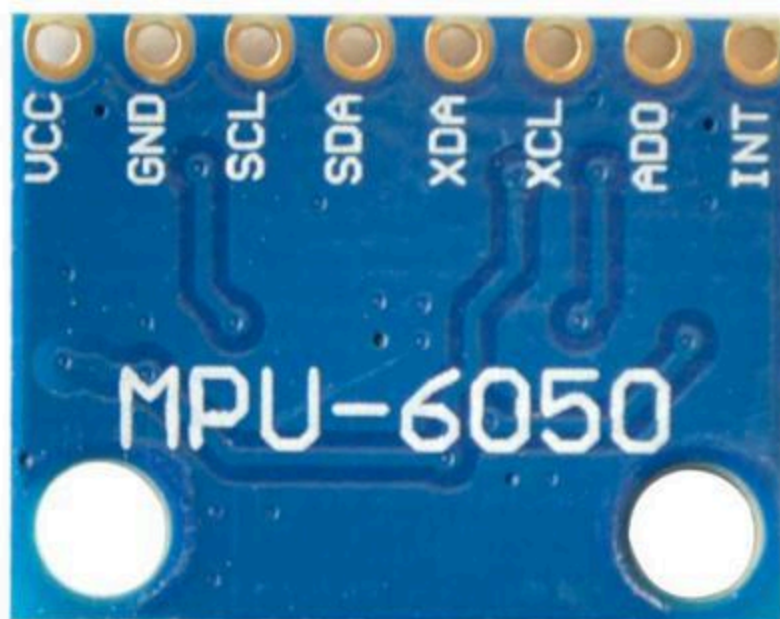
- Embedded C
 - Arduino IDE
-

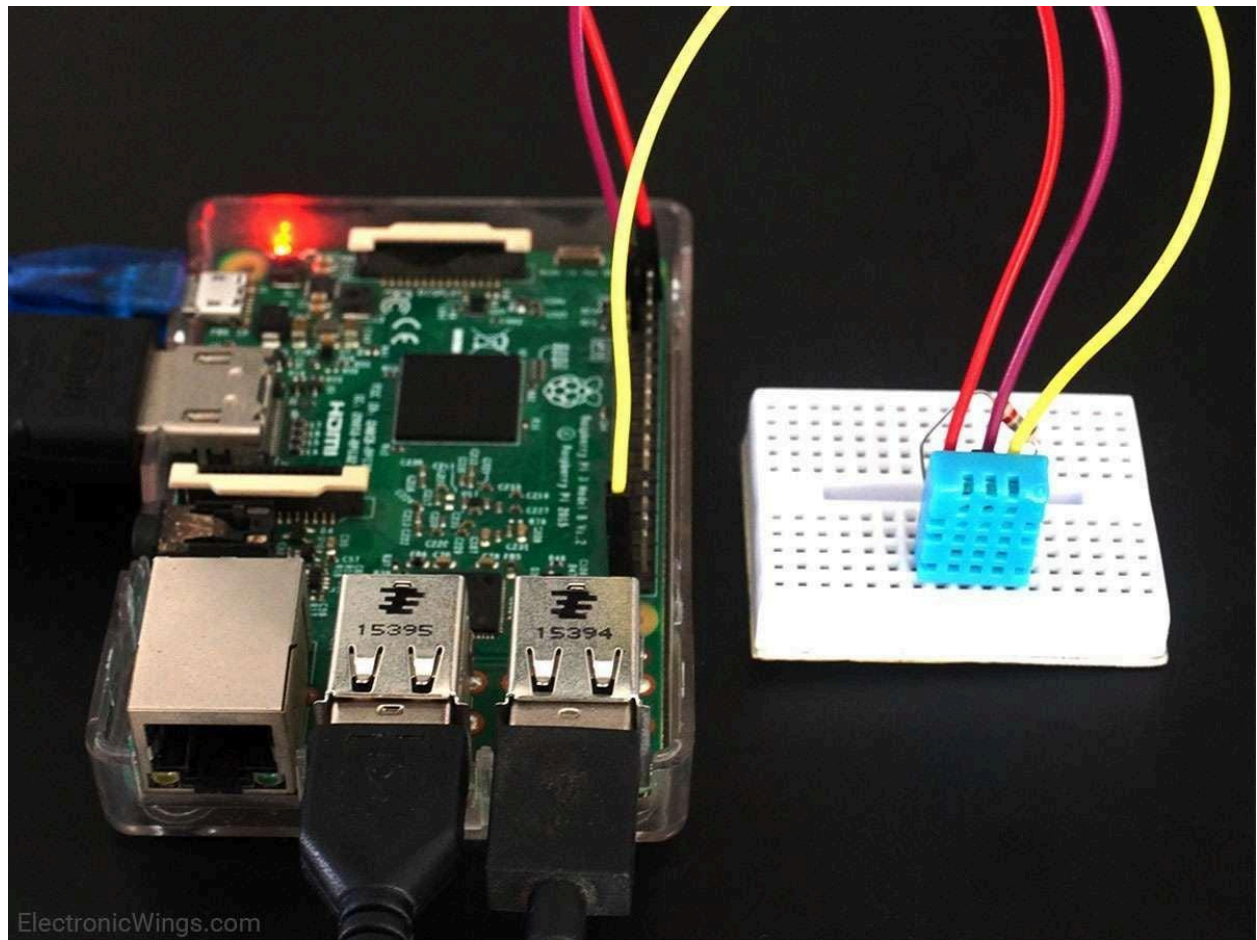
3. Sensors Interfacing

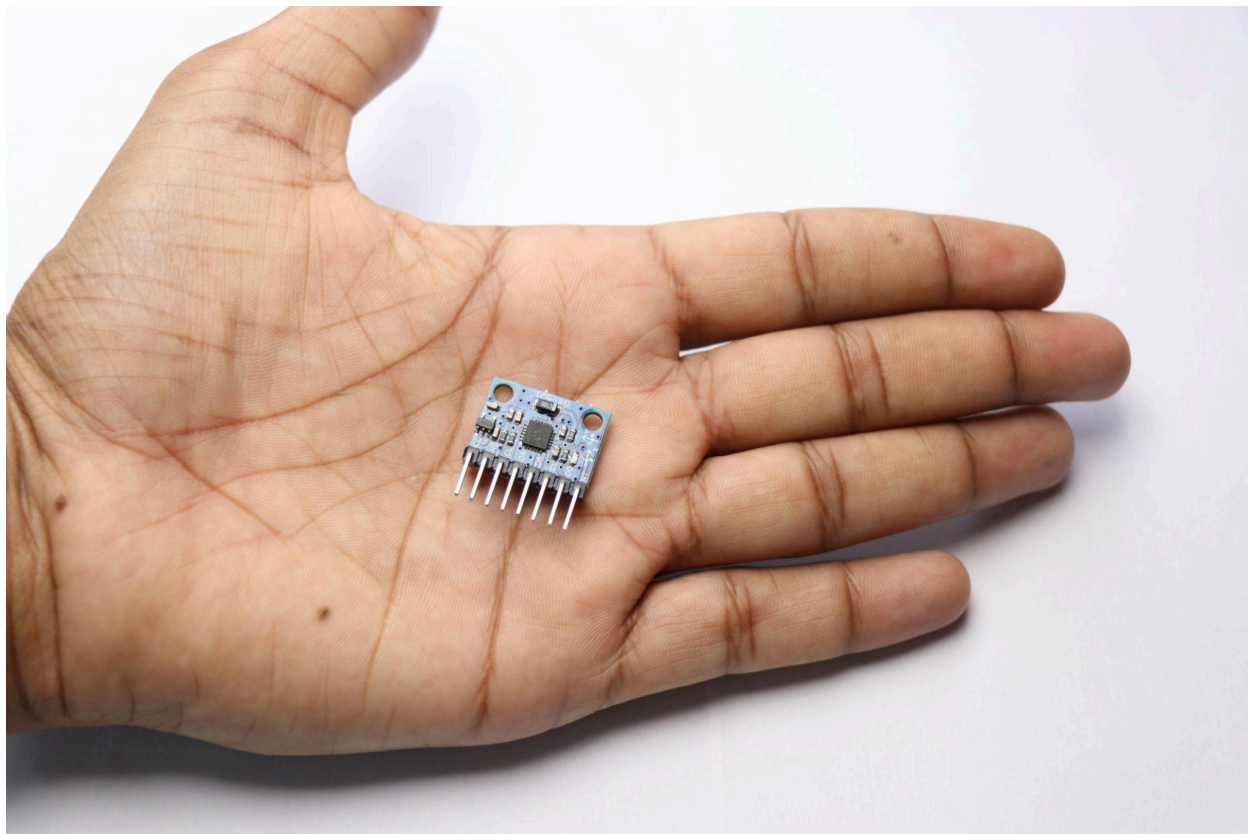


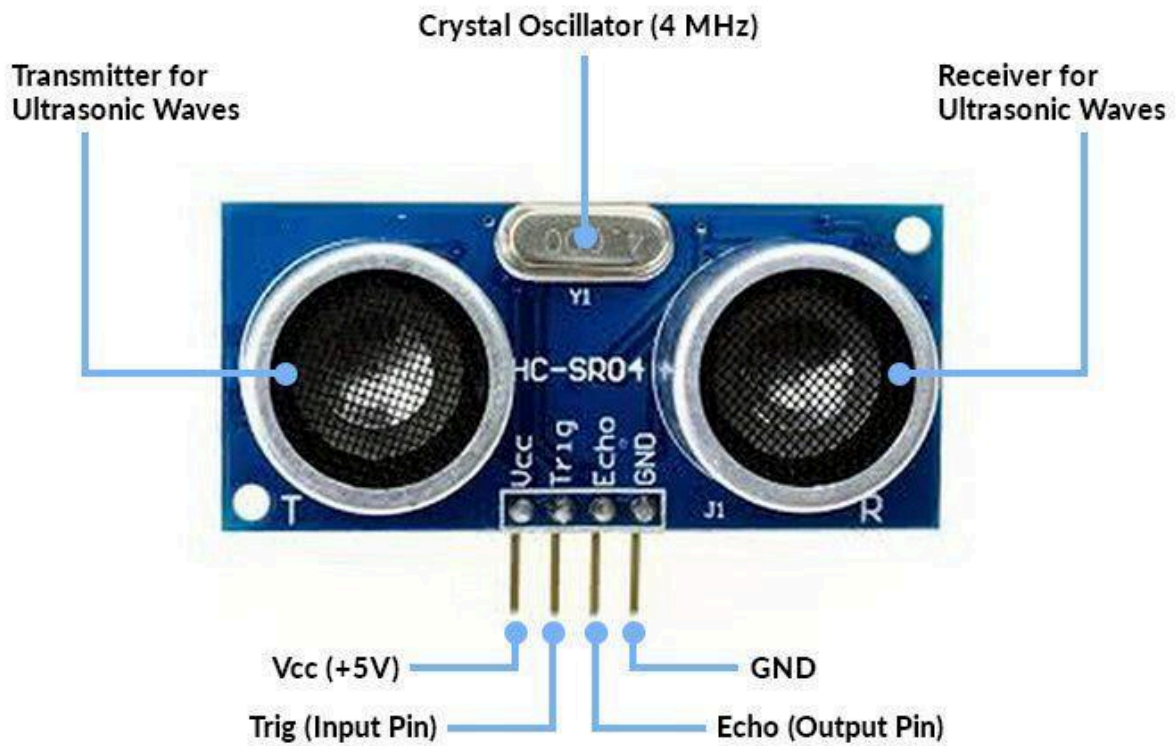












Learn

- I2C
- SPI
- UART
- Analog sensors

- Digital sensors

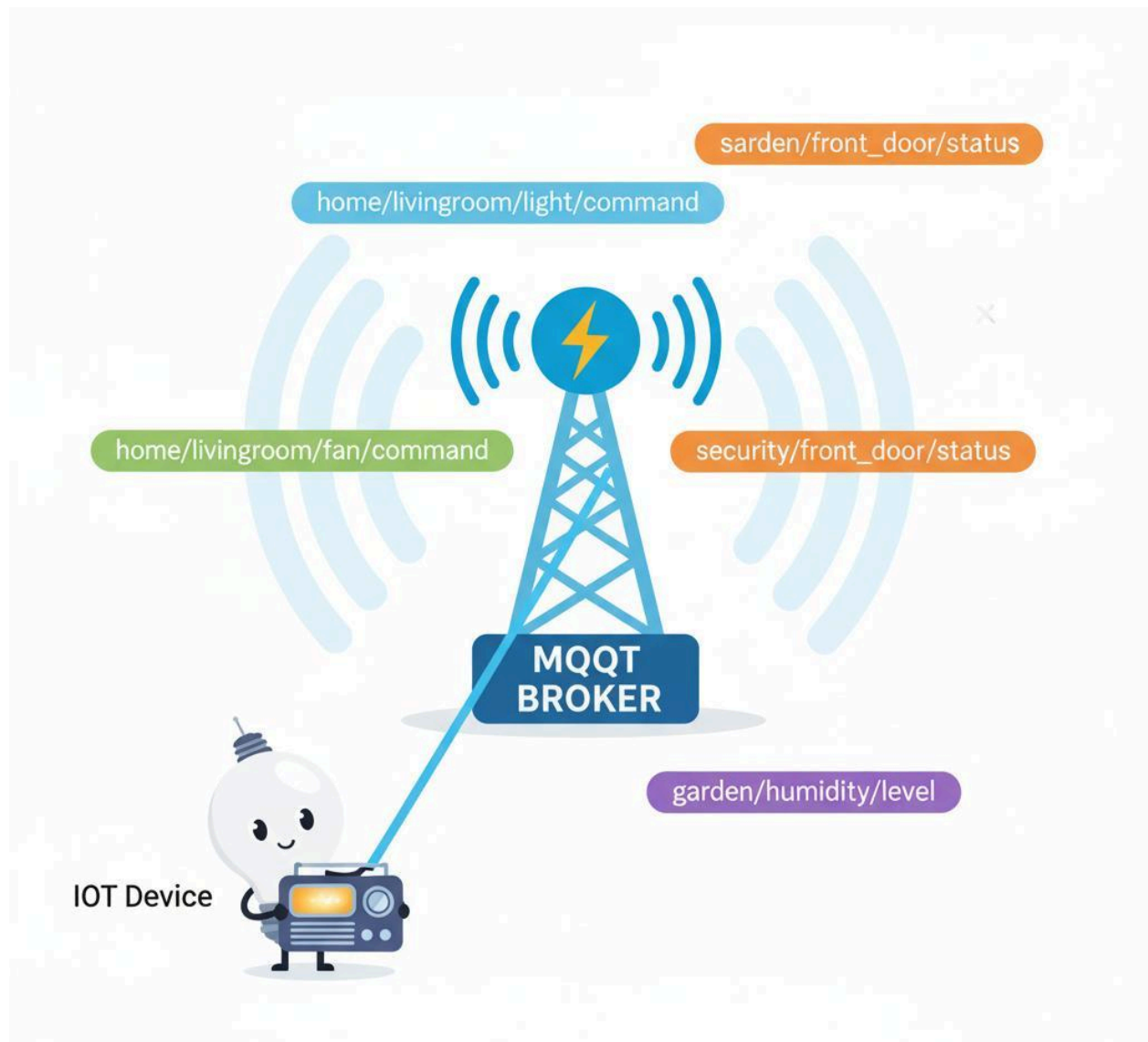
Important Sensors

Sensor	Purpose
DHT11	Temperature
HC-SR04	Distance
PIR	Motion
MPU6050	IMU
GPS	Tracking

Projects

- Obstacle detector
- Temperature monitor
- Motion alarm

4. Wi-Fi + MQTT

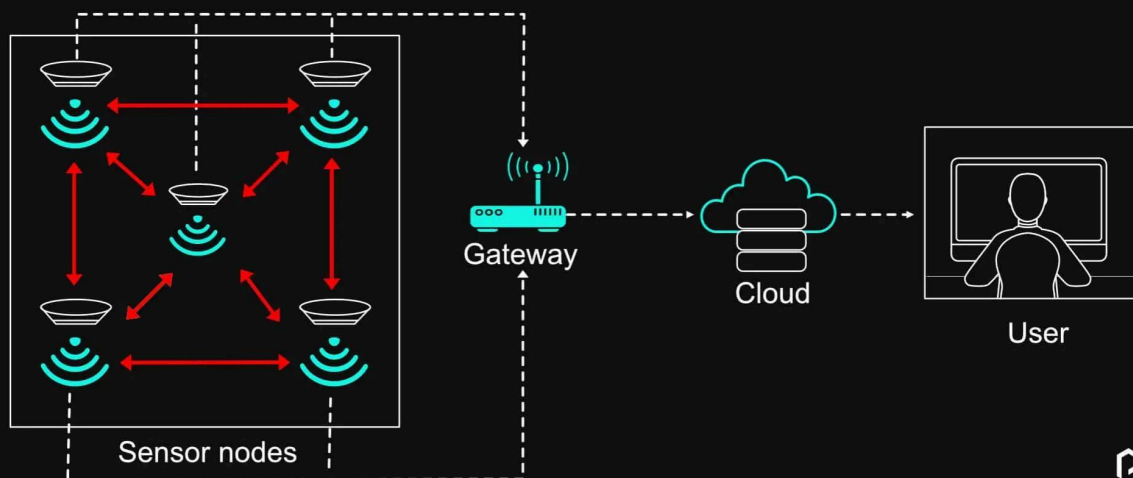


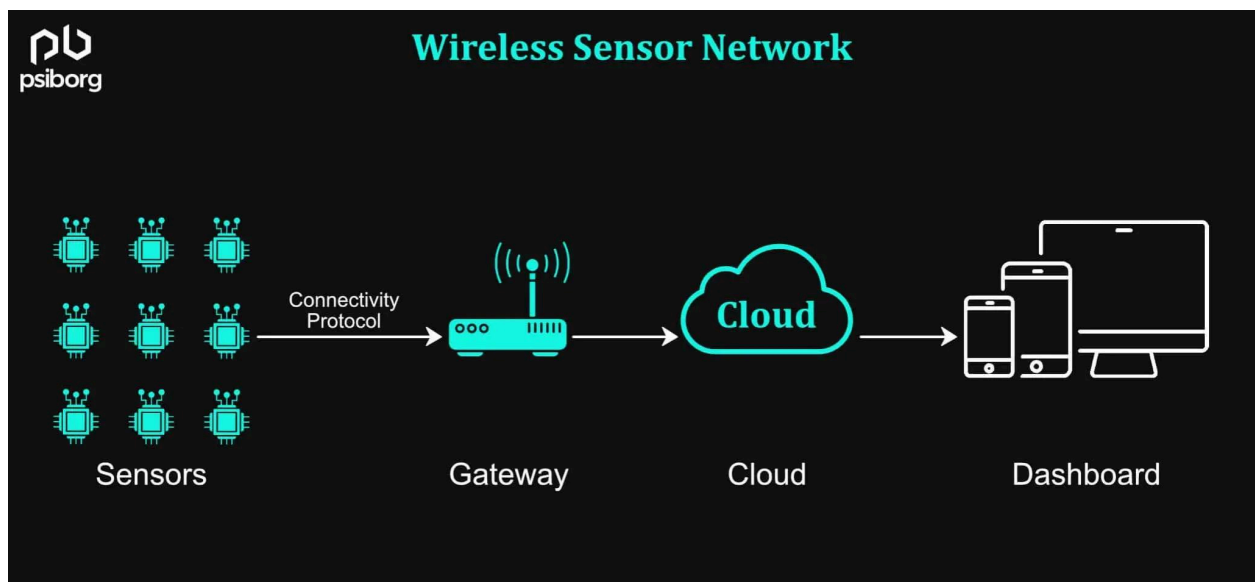
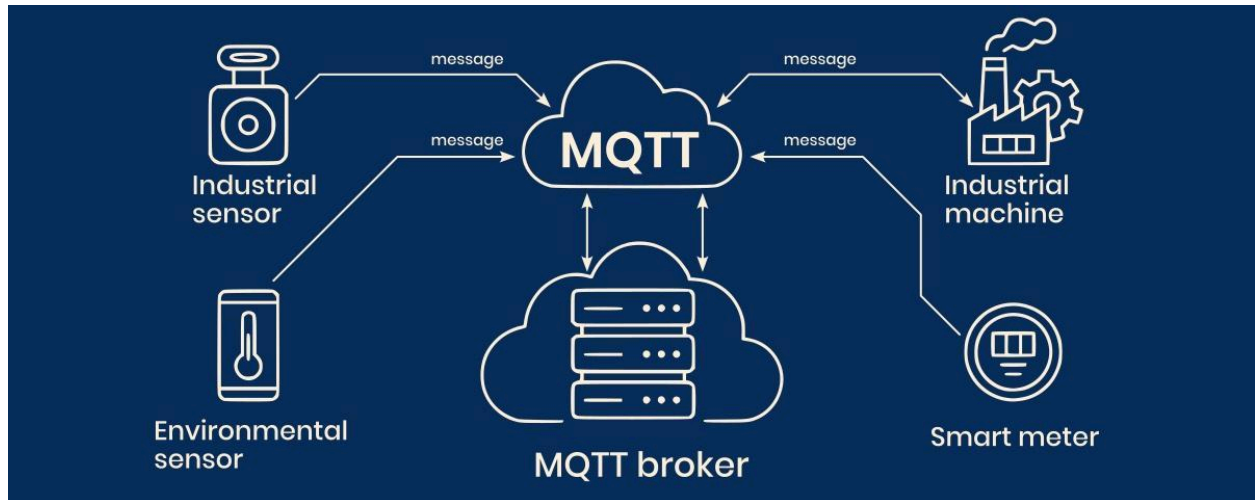


ESP32 Wi-Fi Basics Getting Started



Wireless Sensor Network





Learn

- Wi-Fi communication
- MQTT protocol
- Client/server model
- Real-time data transfer

Platforms

- Mosquitto MQTT
- HiveMQ
- Blynk

Projects

- Remote sensor monitoring
- IoT automation
- Wireless control systems

5. Cloud Dashboards



First



Title of Gauge1



Stuff

numeric ▼ 0 ▲

SLIDE ME

Select an Option ▼

slide 55

Fire ☐

Second

slider1

slider2

slider3

Title of Gauge3

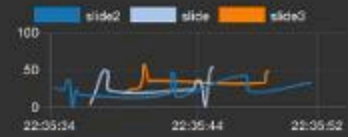


Numeric dropdown Select an Option ▼

Pick Me !

Third

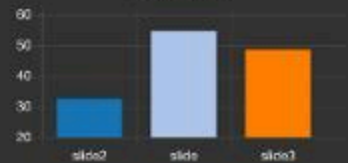
Title of chart



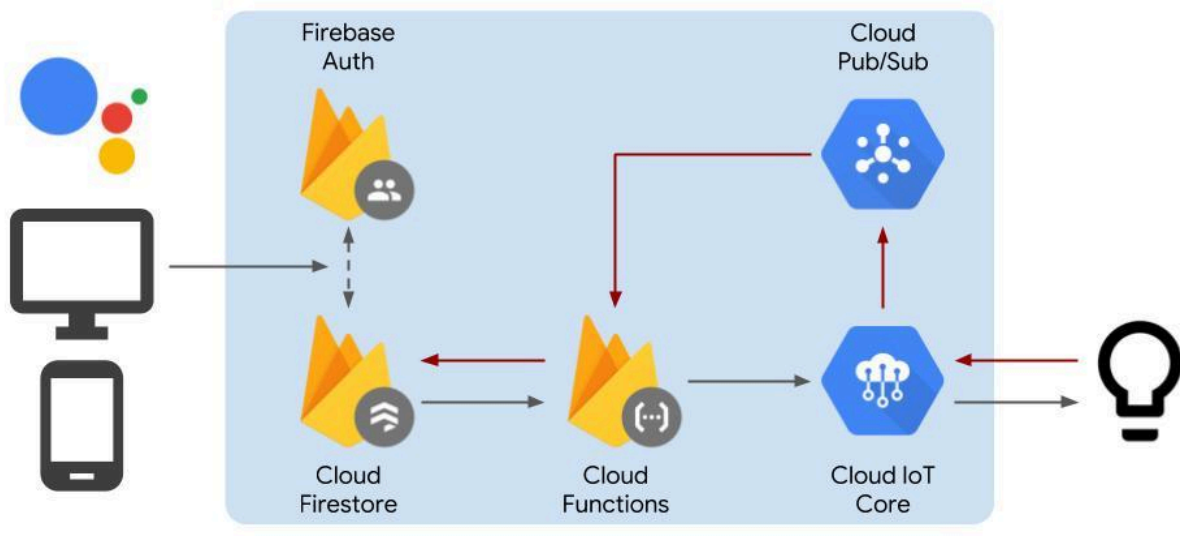
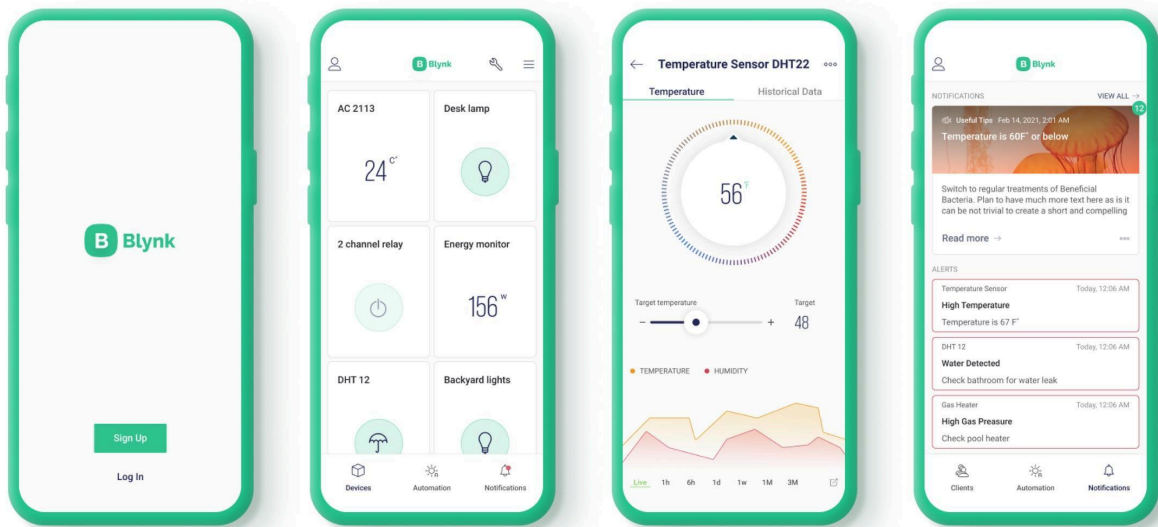
Pie Chart



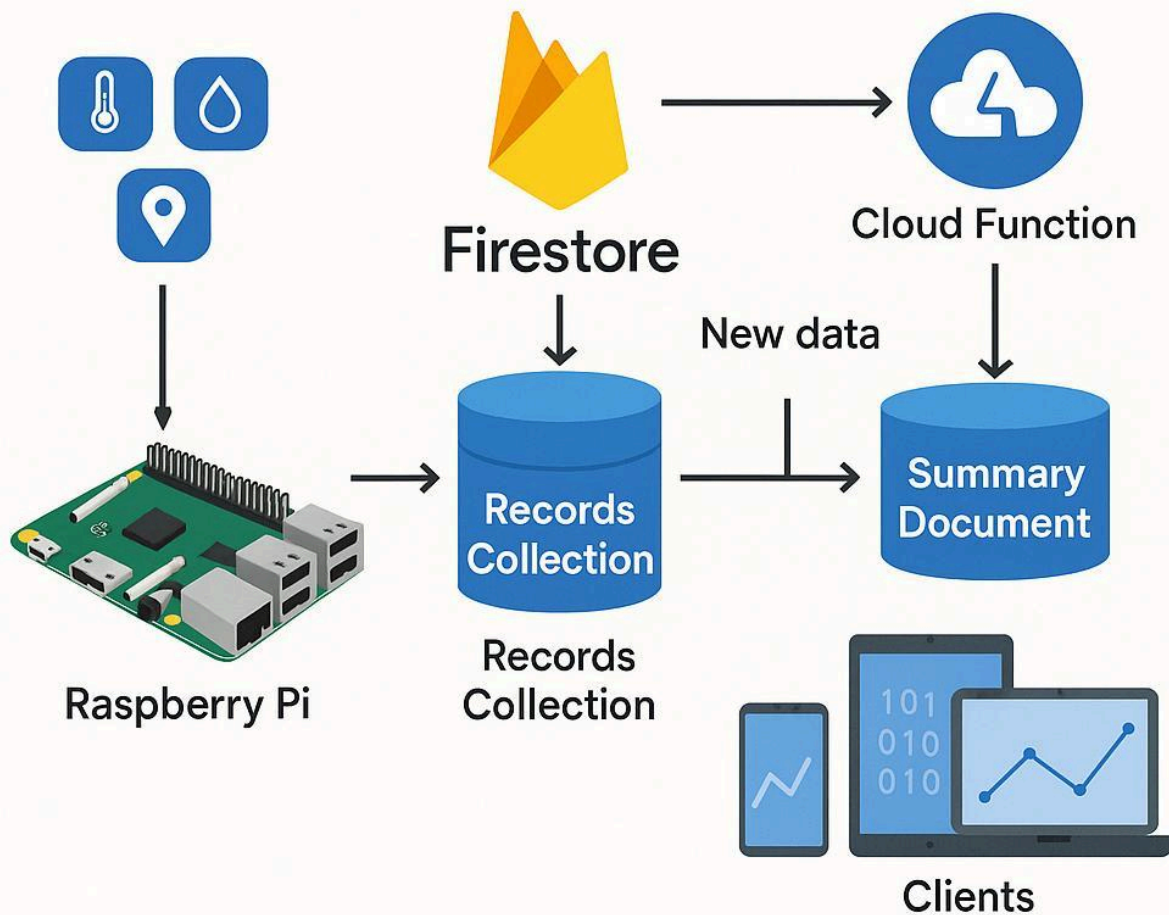
Bar Chart



Horizontal Bar Chart



Real-Time Data Storage and Optimization with Firestore



19:02



Template Name



Power Switch



Temperature

32 °C

Historical Data

● Power ● Temperature ● Datastream 3 ● Datastream 1

Learn

- Cloud storage
- Data visualization
- Real-time monitoring
- Mobile dashboards

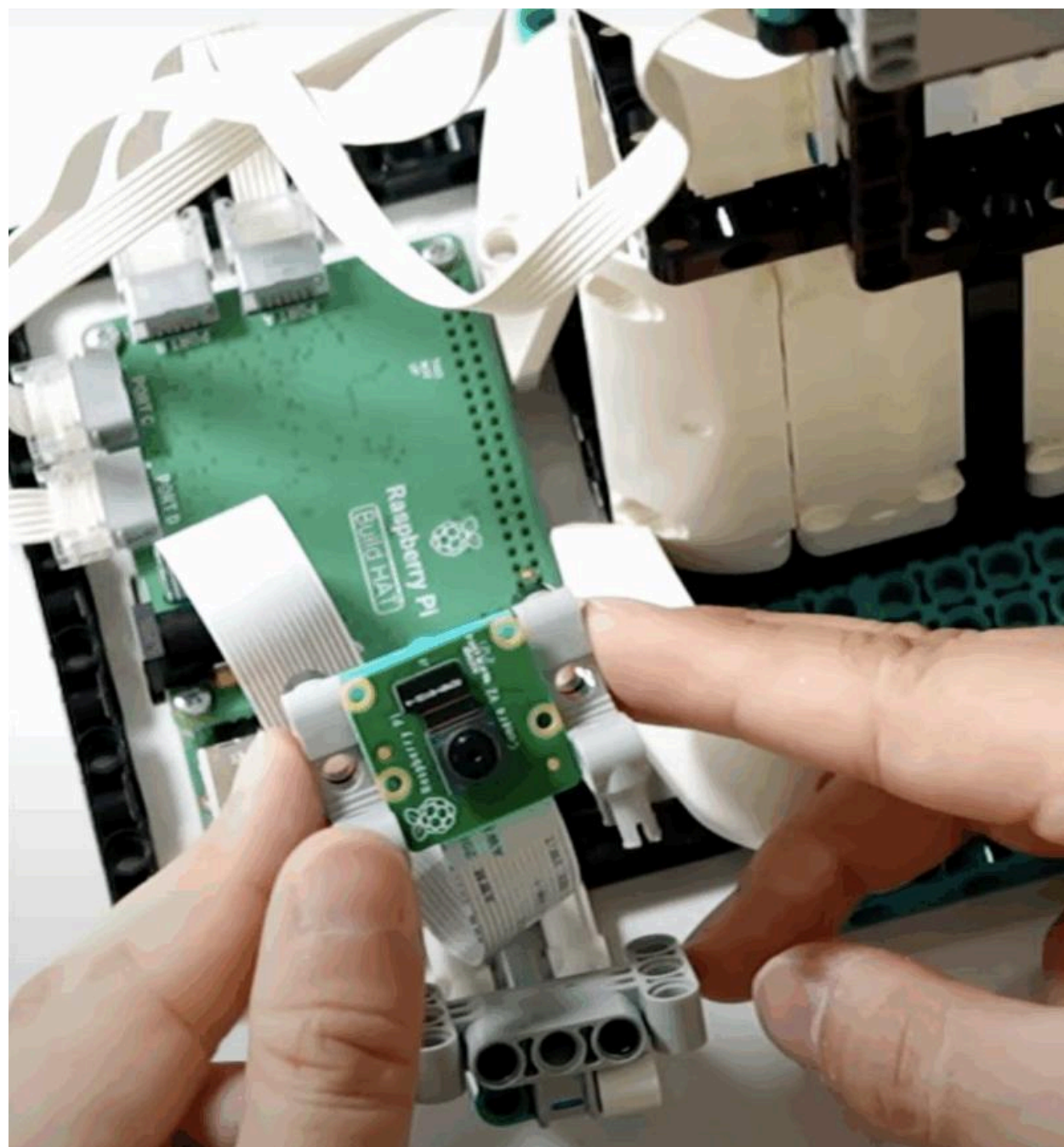
Platforms

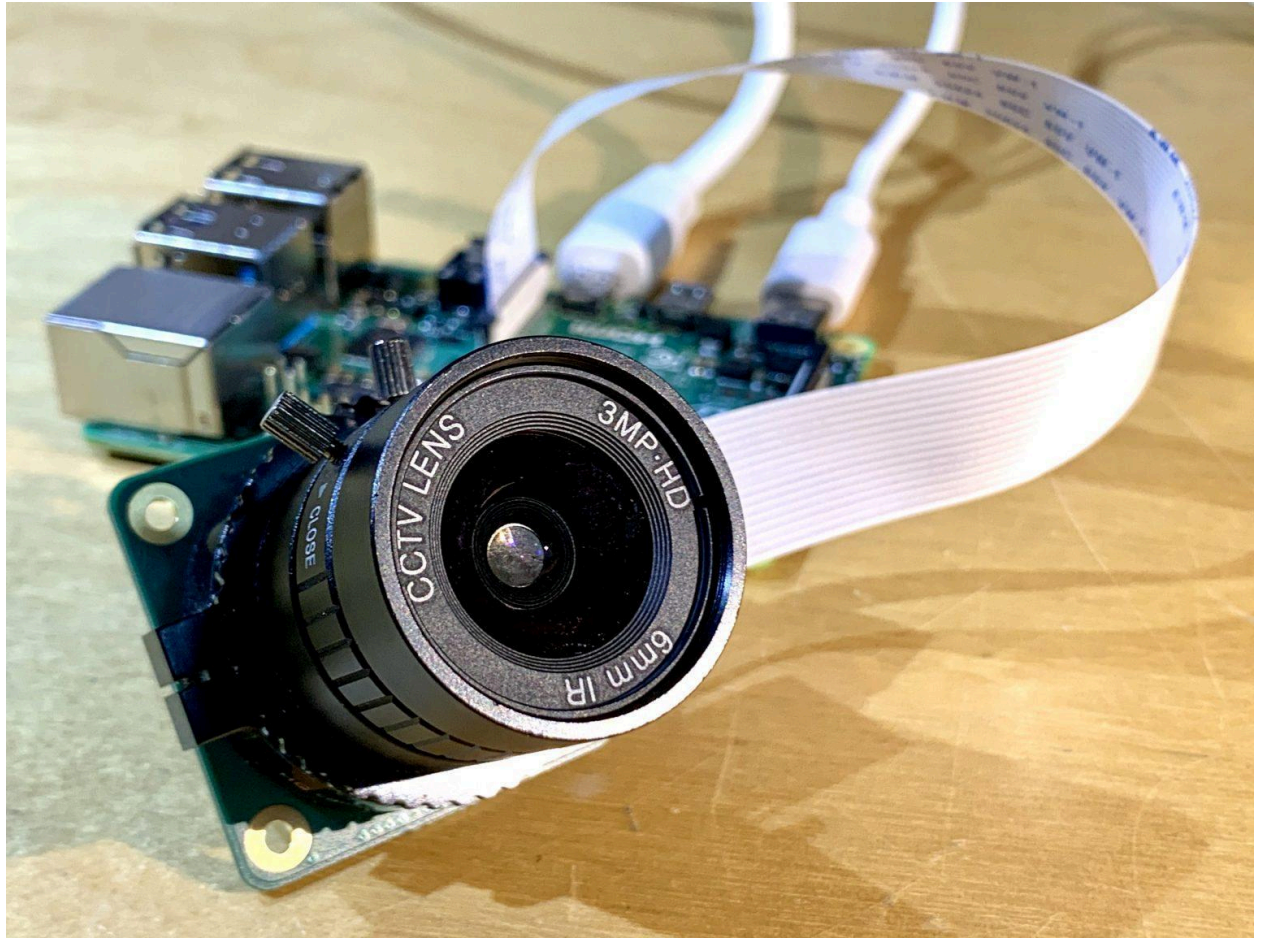
- ThingSpeak
- Firebase
- Node-RED
- Grafana

Projects

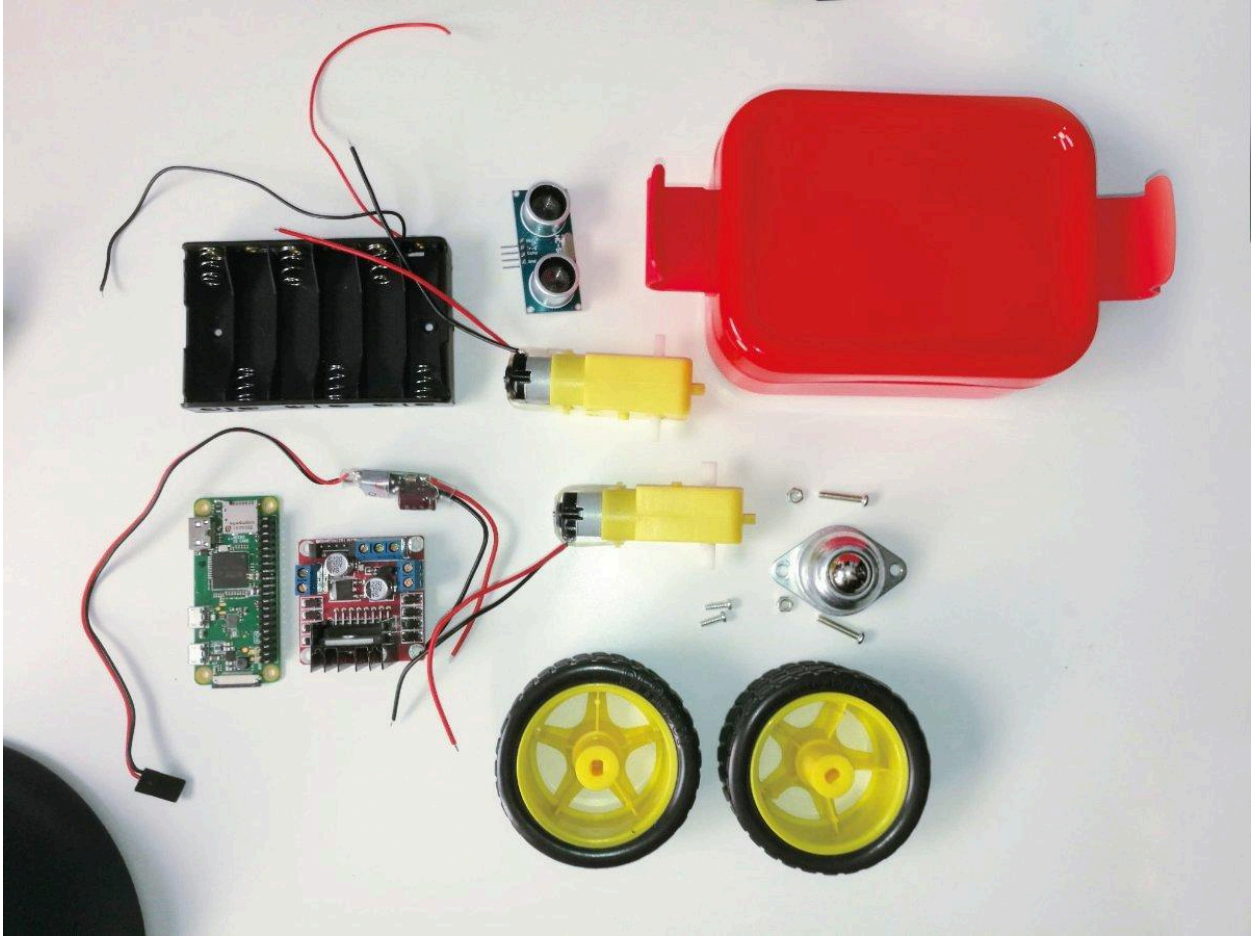
- Smart agriculture dashboard
 - Industrial monitoring
 - Energy monitoring system
-

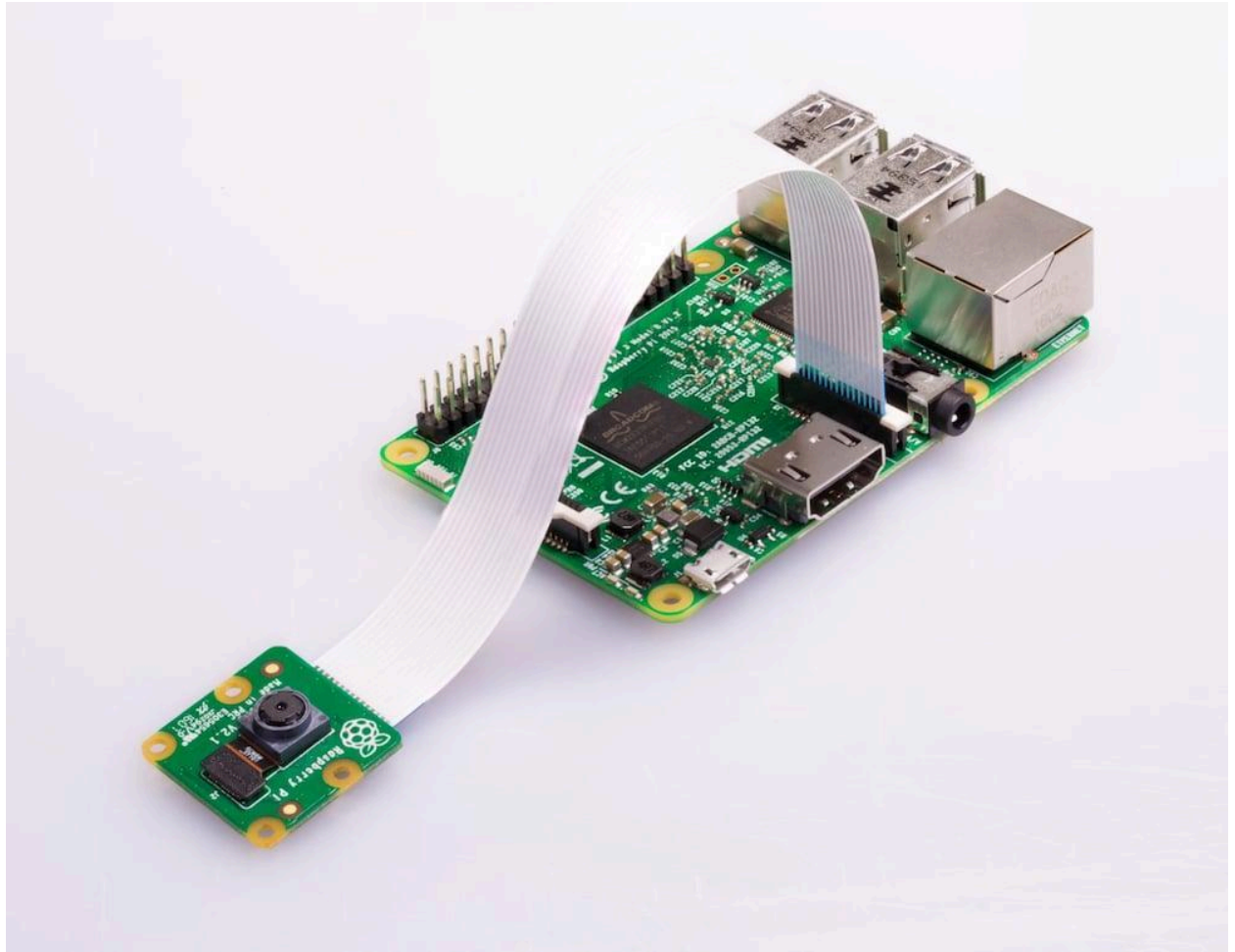
6. Raspberry Pi







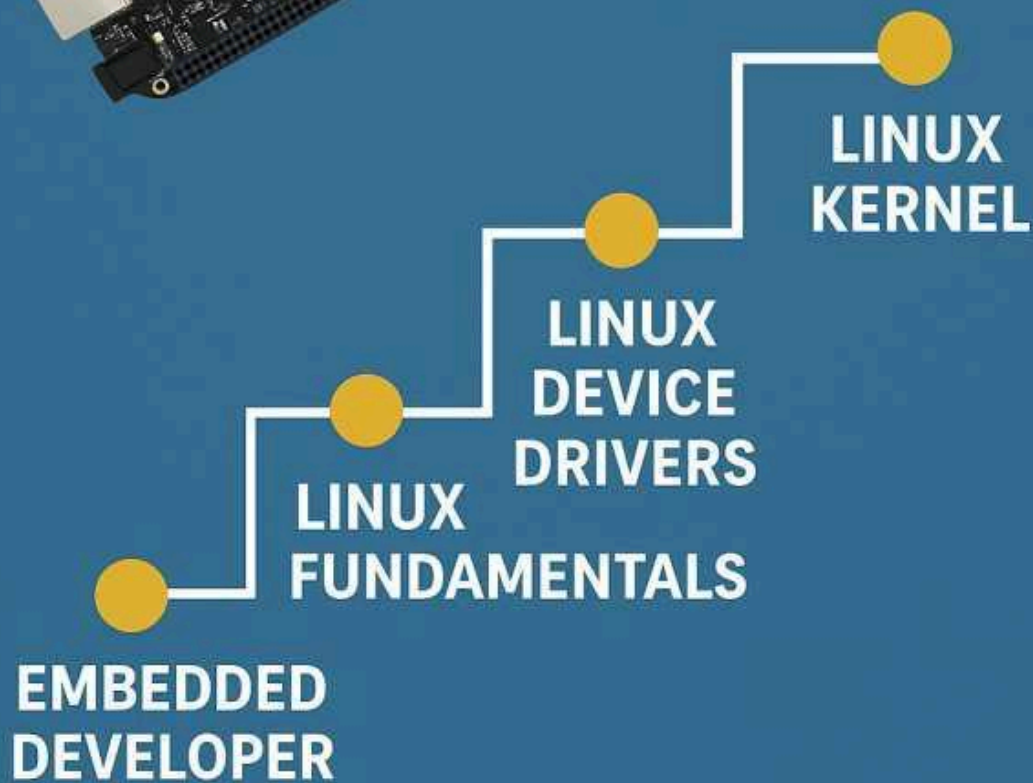
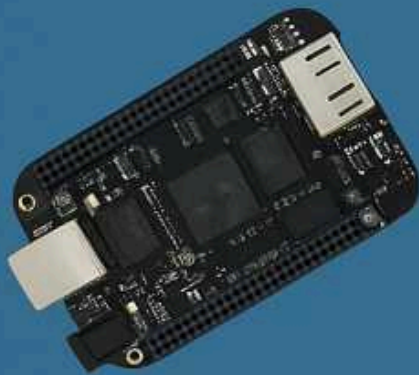




EMBEDDED LINUX



EMBEDDED
LINUX ARCHITECT



Learn

- Linux basics
- Python programming
- Camera interfacing
- Networking

Projects

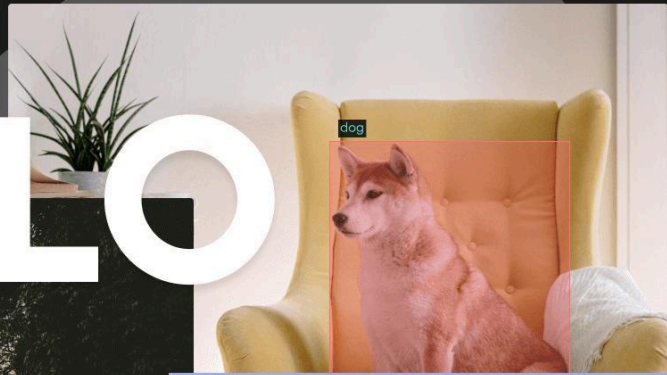
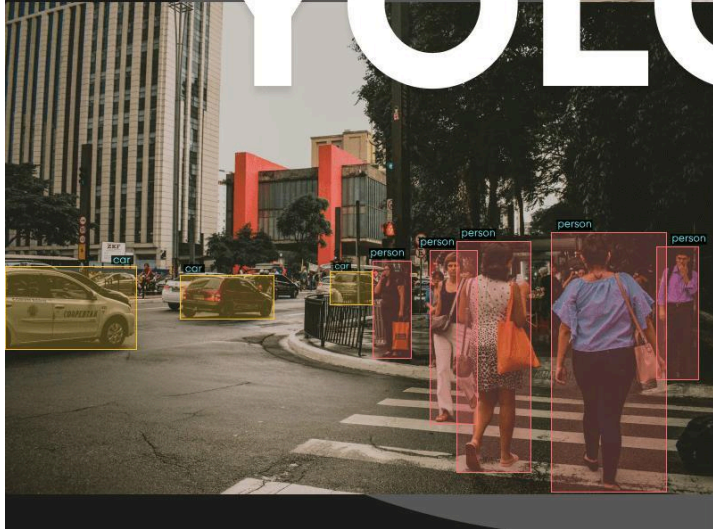
- CCTV system
- Home server
- AI camera

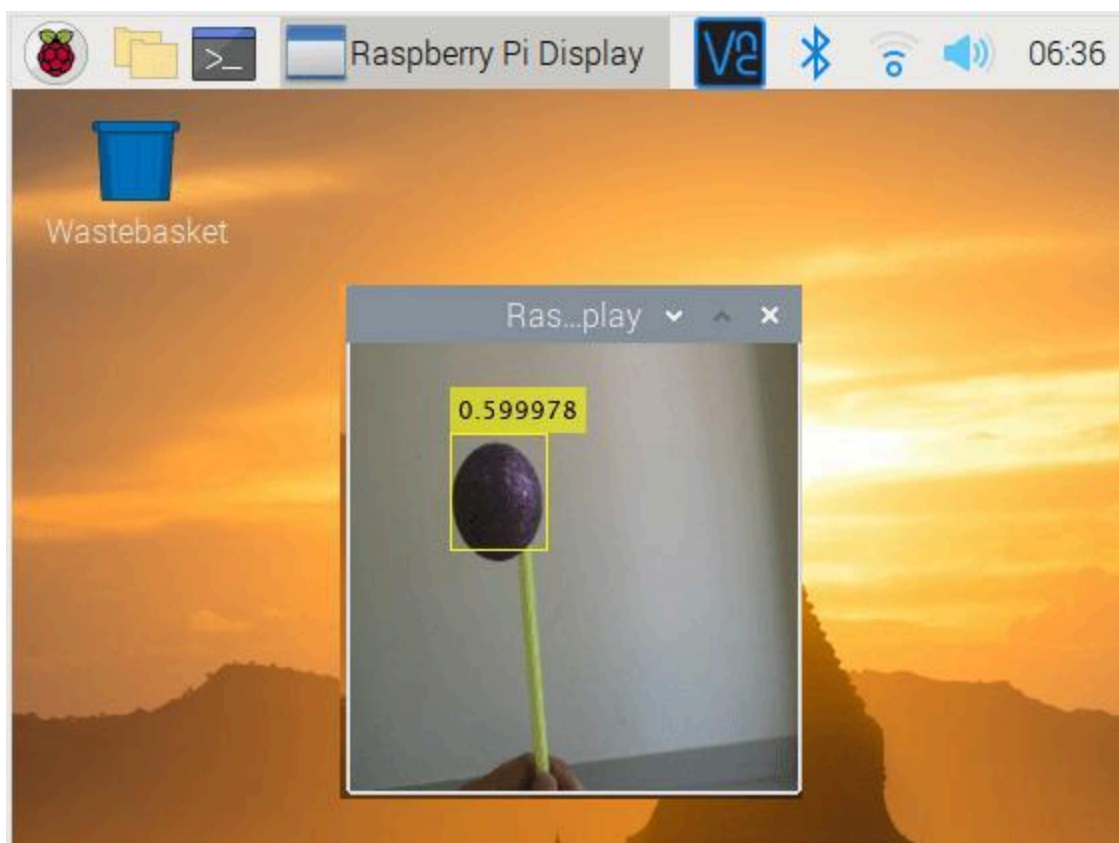
Skills

- SSH
 - Terminal commands
 - Python automation
-

7. OpenCV + AI

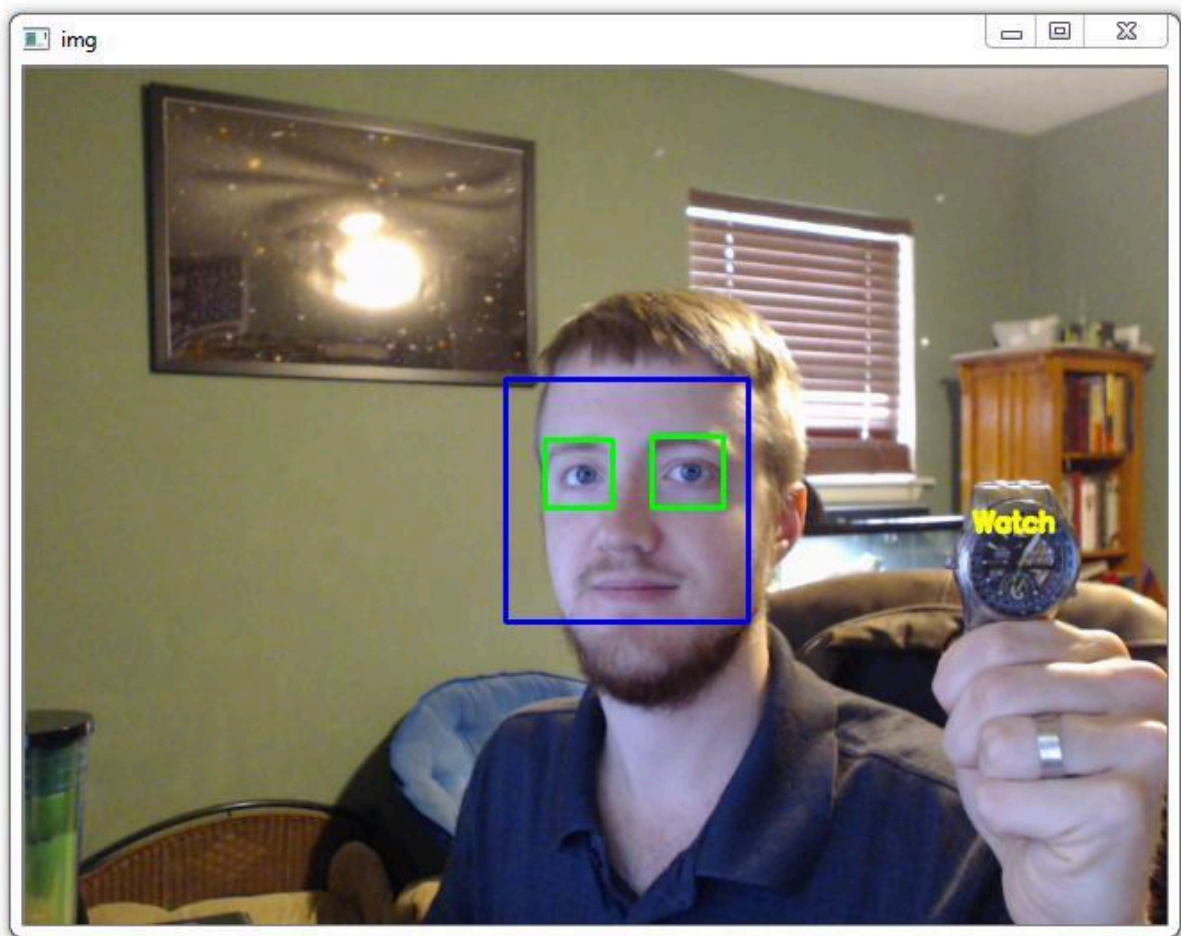
YOLO

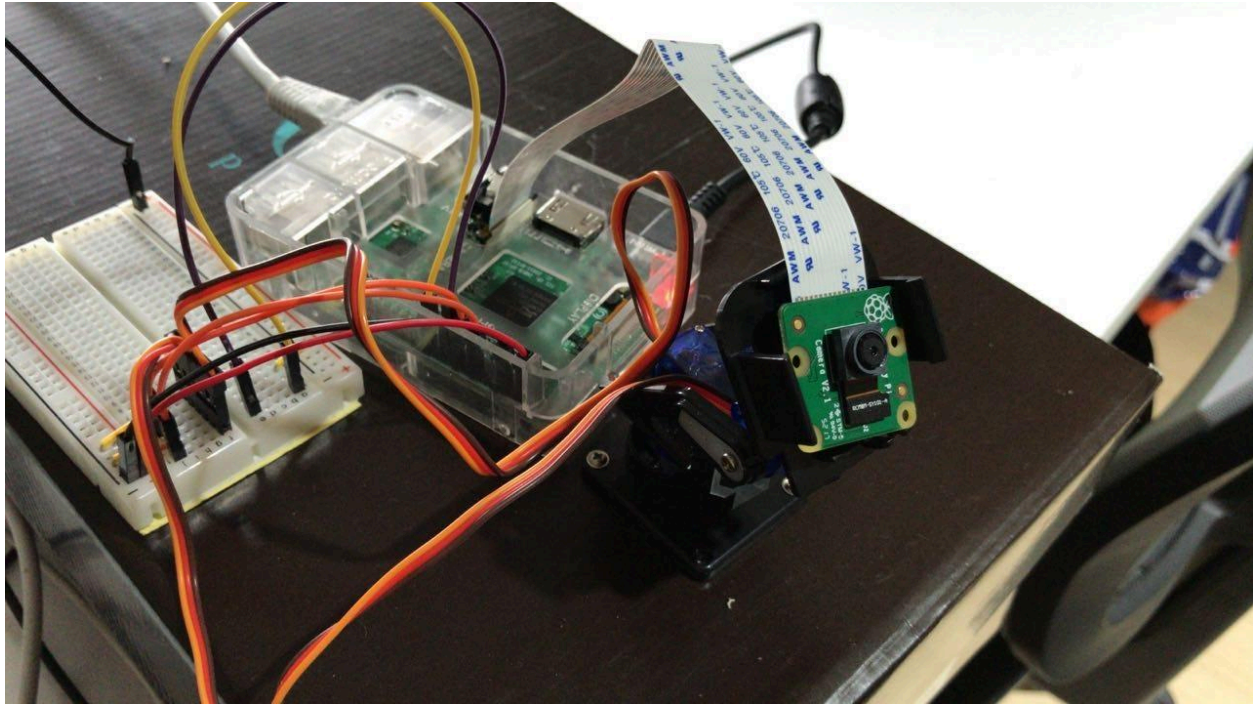












Learn

- Image processing
- Object detection
- Face recognition
- AI models

Technologies

- OpenCV
- YOLO
- TensorFlow
- PyTorch

Projects

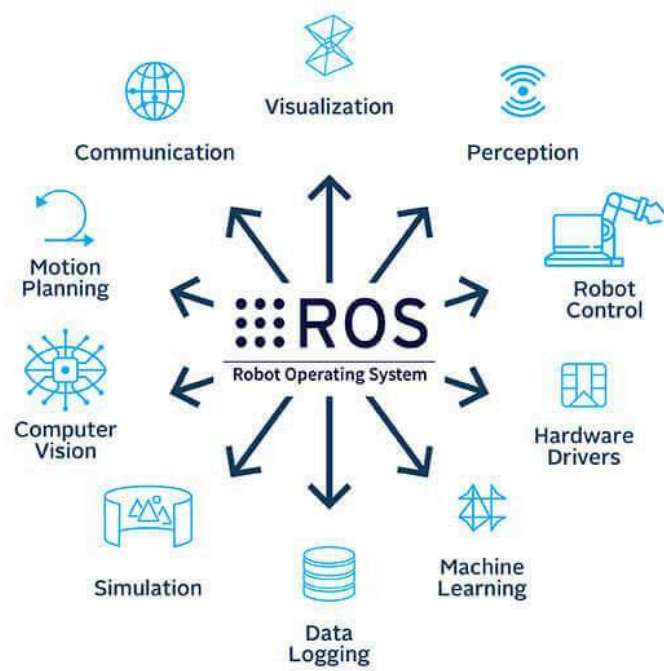
- AI CCTV

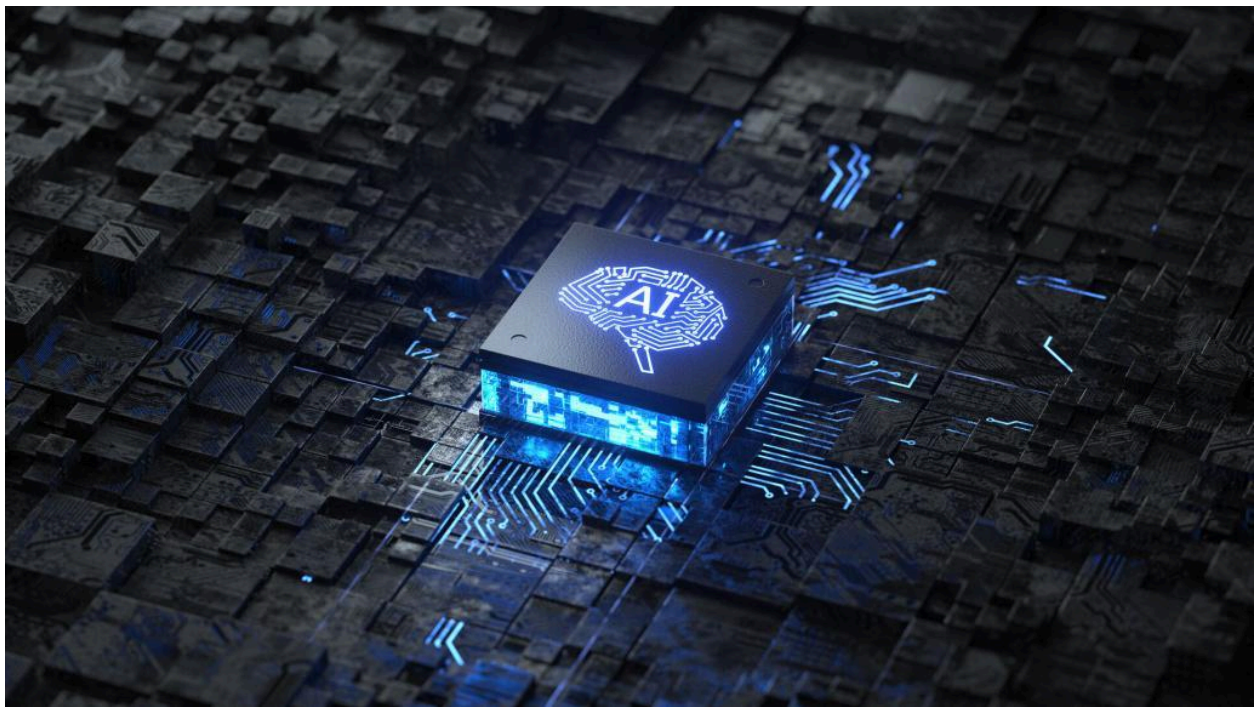
- Thermal detection
 - Smart surveillance
-

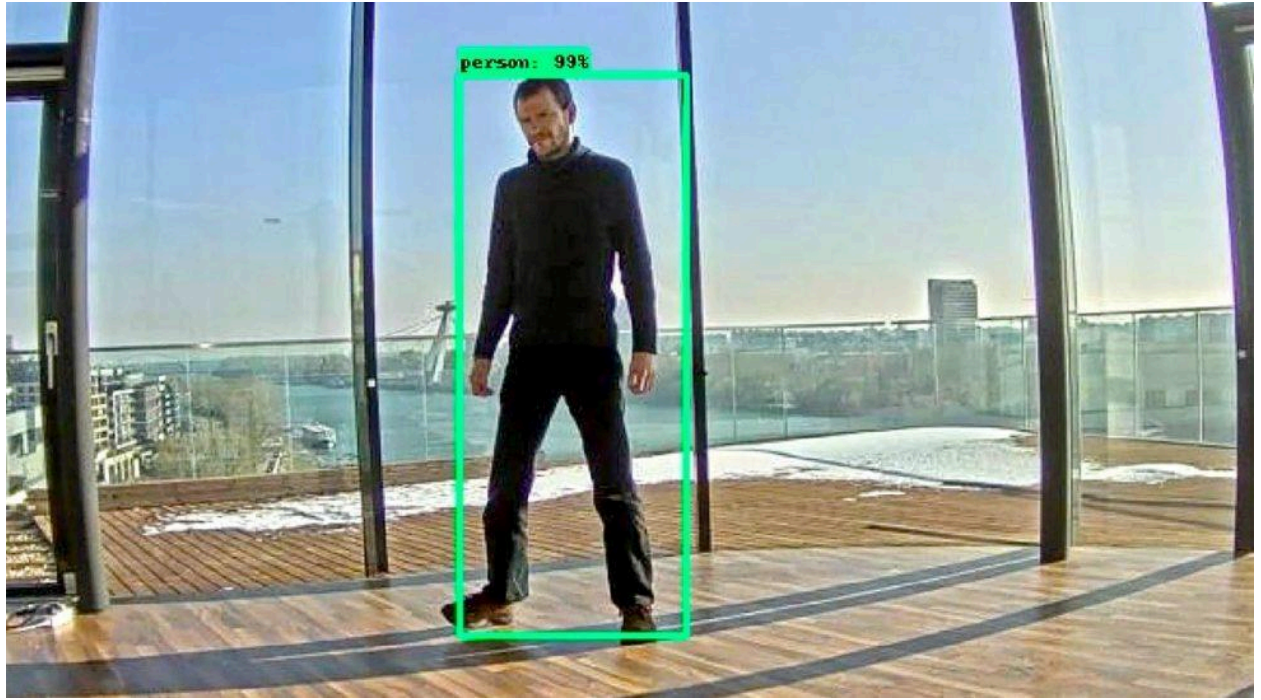
8. Robotics + Automation





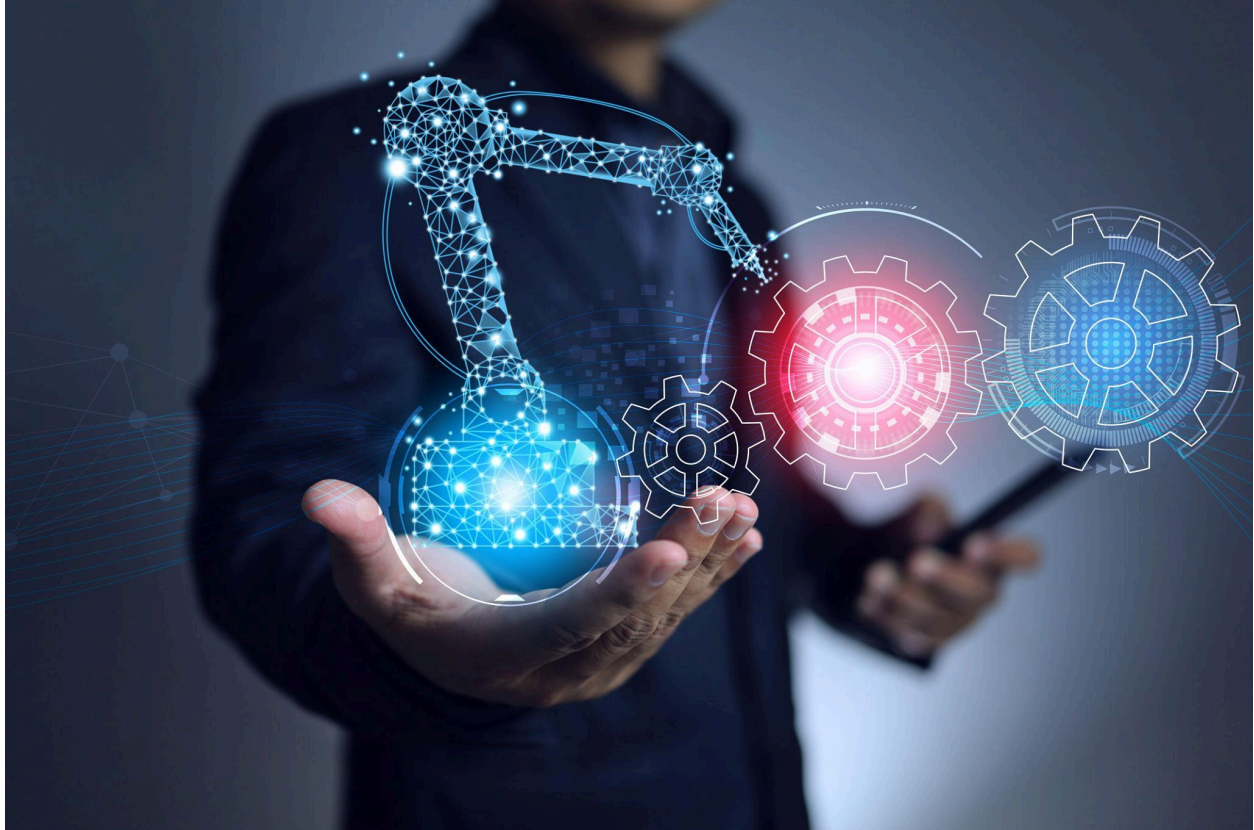












Learn

- Motor control
- Navigation
- ROS
- Autonomous systems

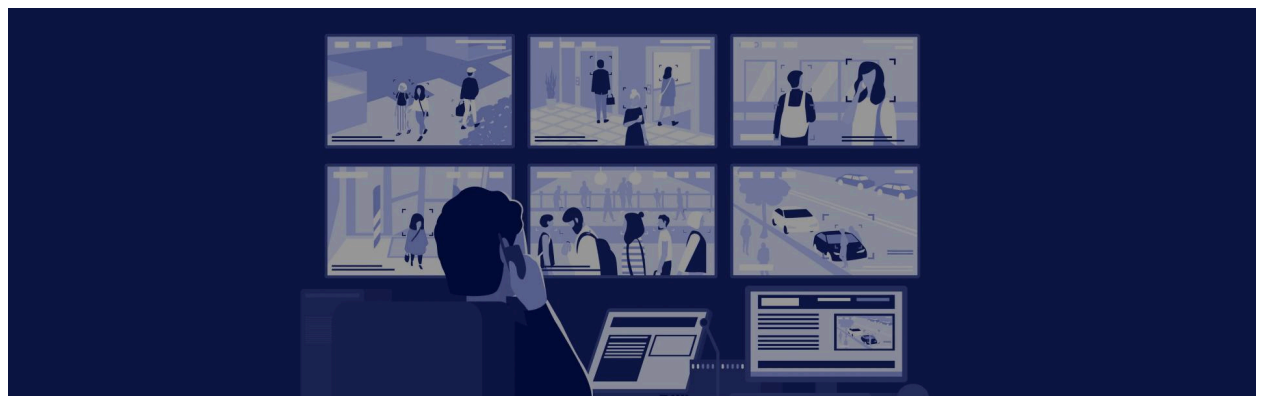
Projects

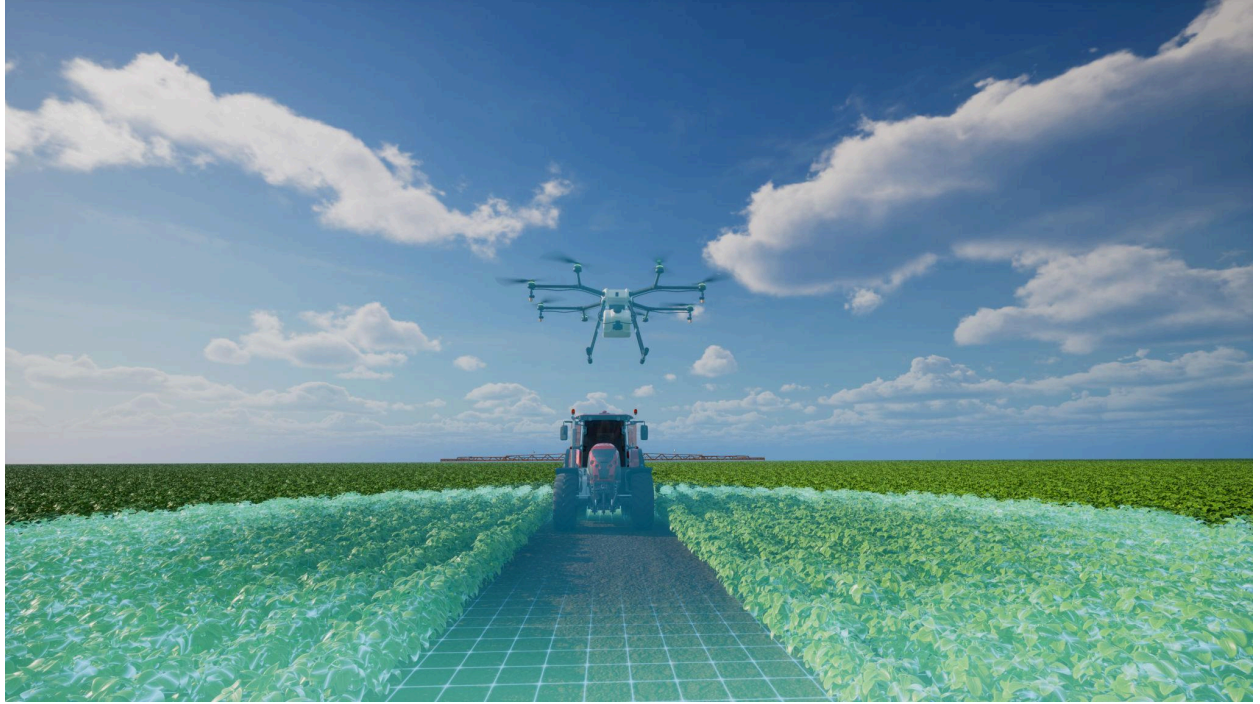
- Autonomous rover
 - Robot arm
 - Drone systems
 - Smart robots
-

Final Goal









After completing this roadmap, you can build:

- AI Surveillance Drone
 - Autonomous Robot
 - Smart Defense System
 - Industrial Automation System
 - Smart City Infrastructure
 - AI IoT Devices
-

Best Future Roles

Role	Technologies
Embedded Engineer	STM32, ESP32
IoT Engineer	MQTT, Cloud
Robotics Engineer	ROS, Sensors
AI Engineer	OpenCV, YOLO
Automation Engineer	PLC, Robotics
Defense Tech Engineer	Thermal + AI + LoRa